FORCED NONINJECTIVITY AND THE SHEET KERNEL AT PRIMITIVE CANONICAL LOCALIZATION SEEDS: A VISUAL EDITION

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ABSTRACT. Let $f\colon U^1 \to U^0$ be an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$. We study Kahn's canonical dynamical localization of the return map $F = f^p$ at the central small filled Julia set $A = K_p$, with its cycle $B = \mathcal{K}_p$ removed. Because F has degree 2^p while its central component has degree 2, the preimage $F^{-1}(A)$ contains excess components; a peripheral loop around one of them lies outside the source localization subgroup and becomes trivial in the target. The localized inclusion is therefore noninjective at every such seed. The complement-cover lift factors canonically as a covering onto an embedded intermediate cover, uniquely within its based commuting class, and at the canonical seed the fiber of the covering carries a group model: the kernel N_{fill} of the source-to-target hole-filling homomorphism. Peripheral winding numbers define a surjection $N_{\mathrm{fill}} \twoheadrightarrow \mathbb{Z}^{2^p-2}$, so every non-singleton fiber is infinite and no finite sheet enumeration is complete. The factorization is preserved, on the complementary locus, by exact pullback base change along Kahn's tower. Applications to Kahn–Lyubich decoration stages require an additional stage-alignment hypothesis and are deferred to a sequel. No rigidity or local-connectivity conclusion is asserted.

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Date: July 27, 2026 Version 1.1 Visual Edition.

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1. Introduction

1.1. **Setting.** Renormalization of quadratic-like maps [DH85], in the generality needed for a priori bounds, takes place in Kahn's category of ψ-polynomial-like maps [Kah23]: pairs (i, f) of a holomorphic *immersion* and a proper branched covering between disks with a common connected filled preimage. A central device of [Kah23], used again in [KL08, KL09], is *localization*: passing to marked covers classified by the fundamental group of a neighborhood of a marked full continuum, so that the dynamics near one small filled Julia set can be studied with the rest of its cycle removed. The *canonical renormalization* of period p is the dynamical localization of the p-th iterate at the central small filled Julia set, with the cycle removed [Kah23, §7.4.3].

The sources are precise about what localization preserves: immersions and branched coverings. The localized inclusion is asserted to be an immersion, never an embedding, and at the unlocalized quadratic-like stage the inclusion $i^p \colon U^p \hookrightarrow U^0$ is literally the inclusion of nested domains [Kah23, §6, p. 31 and footnote 15]. This paper shows that the distinction is necessary in the strongest possible sense.

Throughout, $f\colon U^1 \to U^0$ is an *ordinary* quadratic-like restriction of a quadratic polynomial — ordinary in the sense that the pair is classical rather than ψ-quadratic-like, so that the inclusion is the genuine inclusion of nested domains rather than an immersion — primitively renormalizable of period $p > 1$; we write $F = f^p\colon U^p \to U^0$, $A = K_p$ for the central small filled Julia set, $B = \mathcal{K}_p$ for its cycle, and

$$T' = (U^p \setminus F^{-1}B) \cup A, \qquad T = (U^0 \setminus B) \cup A$$

for the marked surfaces of the canonical localization, with complementary surfaces $\Omega_s = T'\setminus A$ and $\Omega_t = T\setminus A$. A single Jordan disk D_A around A, valid on both sides, determines based localization subgroups $H_s \le \pi_1(\Omega_s)$ and $H_t \le \pi_1(\Omega_t)$ (Definition 2.6), localization covers $\widehat{\Omega}_s$, $\widehat{\Omega}_t$, the complement-cover lift $\widehat{a}$ of i^p, and the completed localized inclusion $\widehat{\imath}\colon T'_A \to T_A$. We call this configuration an *ordinary primitive canonical seed*.

1.2. **Main results.**

Theorem A (Forced noninjectivity; Theorem 4.5). At every ordinary primitive canonical seed of period $p > 1$ the localized inclusion $\widehat{\imath}$ is not injective. Explicitly, there are an excess component $X \neq A$ of $F^{-1}(A)$ and a based loop $\gamma \subset \Omega_s$ with $[\gamma] \notin H_s$ and $(i^p)_*[\gamma] = 1 \in H_t$, and the lift of γ has distinct endpoints identified by $\widehat{\imath}$.

Theorem B (Canonical resolution and kernel model; Theorems 5.5, 5.10 and 6.1). The complement-cover lift factors exactly as $\widehat{a} = e \circ q$, where $q\colon \widehat{\Omega}_s \to P_{\mathrm{ref}}$ is a surjective covering onto the basepoint component of the pullback cover classified by the reflected subgroup $K_{\mathrm{ref}} = (i^p)_*^{-1}(H_t)$, and $e\colon P_{\mathrm{ref}} \hookrightarrow \widehat{\Omega}_t$ is a topological embedding; P_{ref} is the unique connected intermediate cover in its based commuting class on which the remaining leg embeds. Every collision of $\widehat{\imath}$ occurs on the complementary locus and is a sheet identification of q, and the based fiber of q is the pointed coset set K_{ref}/H_s. No extension of the factorization across the filled completion is asserted. At the canonical seed this fiber carries a canonical group model:

$$K_{\mathrm{ref}} = N_{\mathrm{fill}} \rtimes H_s, \qquad N_{\mathrm{fill}} = \ker\big[(i^p)_*\colon \pi_1(\Omega_s) \to \pi_1(\Omega_t)\big],$$

and $n \mapsto nH_s$ is a pointed bijection $N_{\mathrm{fill}} \to K_{\mathrm{ref}}/H_s$. Canonical localization forgets exactly the kernel of the source-to-target hole-filling homomorphism.

Theorem C (Winding quotient and infinitude; Theorem 6.3, Corollary 6.4 and Corollary 5.8). For a quadratic polynomial the degree budget is exact:

$$F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{2^p-2},$$

with every $F|_{X_j}\colon X_j \to A$ a homeomorphism and every X_j disjoint from B. Peripheral winding numbers about the X_j define a surjective homomorphism

$$W\colon N_{\mathrm{fill}} \twoheadrightarrow \mathbb{Z}^{\,2^p-2},$$

so $\operatorname{rank} H_1(N_{\mathrm{fill}};\mathbb{Z}) \geq 2^p - 2$. In particular every non-singleton fiber of $\widehat{\imath}$ (equivalently, every fiber over the embedded resolved complement) is infinite, and no finite sheet enumeration of the canonical localization is complete.

The count $2^p - 2$ itself is an elementary consequence of the degree budget; the content of Theorem C is the kernel identification and the peripheral-winding surjection. The factorization of Theorem B is standard covering-space theory; the contribution is the identification of the actual dynamical subgroup defect and of its canonical kernel model, not the invention of intermediate covers. The factorization and its fiber data are moreover stable: they are preserved, on the complementary locus, by exact pullback base change along every stage of Kahn's tower (Section 7).

1.3. **Method.** The proof of Theorem A runs along a five-step spine. The degree budget ($\deg F = 2^p$ against central degree 2) produces an excess component X of $F^{-1}(A)$; primitivity places X outside B, so X lies in the set removed from the source but not from the target; peripheral isolation puts an isolating Jordan disk around X; the winding number about X defines a homomorphism vanishing on H_s but not on the peripheral class, so the class lies outside H_s; and the target contracts the peripheral loop across the isolating disk. The strict subgroup defect $H_s \subsetneq (i^p)_*^{-1}(H_t)$ then yields an actual point collision by the covering-space endpoint criterion (Theorem 3.1).

1.4. **Relation to the literature.** Concerning injectivity, the primary sources assert preservation of immersions, not of embeddings. The most recent treatment, by Kahn, Kapiamba and Lyubich — including both authors of the original localization construction — still asserts only the existence of the lifted ψ-quadratic-like map, with no global injectivity statement [KKL26, §§2.2–2.3], and the current survey of the surrounding program does not address injectivity of localization [Dud25]. We are not aware of any statement in the literature determining whether the canonically localized inclusion is injective.

The closest published anticipation is by the source authors. In the proof of [KL08, Lemma 3.4(ii)] (p. 68 of the journal pagination), the component index map for the pseudo-piece immersion is described as "surjective but not necessarily injective." Theorem A differs in three ways: it is a forcing statement rather than a possibility statement; it concerns the localized inclusion itself rather than a component index map (noninjectivity of the index map does not imply noninjectivity of the glued immersion, since distinct universal-cover tiles can occupy disjoint sheets over one target tile); and it concerns Kahn's localization functor at a canonical seed rather than Kahn–Lyubich pseudo-puzzle tiles. Separately, the structural shape appearing in Theorem B — an immersion that is a covering onto an embedded image away from a proper compactum — has been axiomatized in recent work of Dudko, Luo and Lyubich [DLL25, §2.1] in the Siegel setting; there it is

imposed as a definition, whereas here it is derived at the canonical seed. Finally, we note that in Kahn's own path model of the canonical localization [Kah23, §7.4.5] the failure of injectivity is a short argument once one looks for it; to our knowledge no one has recorded it.

1.5. **Scope.** These results contradict nothing stated in [Kah23, KL08]: the sources assert immersion preservation, an embedded inner collar, and the forward direction of path equivalence, and all three are compatible with Theorems A–C. Remark 4.7 gives the precise sheet-level reconciliation: the collision joins distinct lifted sheets over the same physical basepoint and is not a self-collision of the distinguished embedded local sheet. What is refuted is any *extension* of the sources that treats canonical localization as embedding-preserving. The results are corrective with respect to any larger program: the canonical localized object at an ordinary primitive seed is not an embedded piece but an embedded cover together with an infinite coset-valued sheet fibration and its transport, and any construction that uses the localization must either carry that fibration or prove a quotient theorem. Applications to Kahn–Lyubich decoration stages require an explicit stage-alignment hypothesis; they are developed in a sequel and none of Theorems A–C depends on them. No a priori bounds, rigidity, or local-connectivity conclusion is asserted or implied.

1.6. **Organization and conventions.** Section 2 collects the definitions and the precise consequences of the sources that are used, with published citations. Section 3 proves the subgroup-reflection criterion and the peripheral-isolation toolkit. Section 4 proves Theorem A; Section 5 constructs the canonical resolution; Section 6 proves the kernel model and the winding quotient; Section 7 proves exact base-change stability. Section 8 states the operational consequences and open questions. Appendix A presents two abstract counterexamples showing that no argument from the sources' stated hypotheses alone can restore injectivity; their exact computational witnesses are in the ancillary files.

For paths α, β with $\alpha(1) = \beta(0)$, the concatenation $\alpha * \beta$ traverses α first. Kahn I refers to arXiv:math/0609045v3 (28 December 2023); KL II refers to arXiv:math/0609046v2, published as Ann. Sci. Éc. Norm. Supér. (4) **41** (2008) 57–84. A source concordance, archive hashes, and the exact computational witnesses are distributed with the manuscript sources.

Disclosure. Research, source comparison, drafting, verification support, and the exact computational witnesses of Appendix A used Fae Build, Anthropic Fable 5, and OpenAI GPT-5.6 Sol. The author checked the source identifications against the cited published versions; independent specialist confirmation has not yet been obtained. No model output is treated as mathematical authority, and the author takes full responsibility for every claim.

2. Canonical localization at a primitive seed

This section fixes the objects used throughout and records, with published citations, exactly which statements of the sources enter the arguments. The last three items — Lemma 2.5, Definition 2.6 and Lemma 2.7 — are elementary and are not in the sources; they legitimize the Jordan local neighborhood, fix based representatives of the localization subgroups valid simultaneously on both sides of the canonical seed, and reduce collisions of the completed localized inclusion to collisions of the lift between complement covers.

2.1. ψ-polynomial-like maps and localization.

Definition 2.1 (FJ-sets and ψ-polynomial-like maps)**.** A subset of $\mathbb{C}$, or of a Riemann-surface disk, is an *FJ-set* if it is compact, connected, full, and has more than one point. A *ψ-polynomial-like map* of degree d with connected Julia set is a pair of holomorphic maps $(i, f)\colon \mathbf{U}' \to \mathbf{U}$ between disks, together with an FJ-set $K \subset \mathbf{U}$, such that i is an immersion, f is a proper branched covering of degree d,

$$i^{-1}(K) = f^{-1}(K) =: K',$$

and K' is connected [Kah23, Definition 7.1, p. 36]. One calls K the *filled Julia set* of (i, f). The source records that K is determined by the pair, that K' is again an FJ-set, that connectivity of K' is equivalent to all branch values of f lying in K, and that i restricts to a bijection $K' \to K$ [Kah23, Definition 7.1 and the remark following it, p. 36], [Kah23, §7.3.2, p. 42]. A ψ-polynomial-like map of degree two is *ψ-quadratic-like*.

Definition 2.2 (Marked surfaces, local pairs, localization)**.** A *marked Riemann surface* is a pair (S, A) with S a Riemann surface and $A \subset S$ compact, connected, and full in the sense that no component of $S \setminus A$ has compact closure in S; a *morphism* $g\colon (S, A) \to (R, C)$ is a holomorphic map with $A = g^{-1}(C)$. The pair (S, A) is *local* if every component of $S \setminus A$ is an annulus, in which case S deformation-retracts into every neighborhood of A; a *local neighborhood* of A is a subsurface $U \supset A$ with (U, A) local, and every neighborhood of A contains one. A morphism is a *local cover* if it restricts to a bijection of the marked sets and to a covering map of their complements. The *localization* of (S, A) is the local marked surface $(S_A, \widehat{A})$, unique up to isomorphism, carrying a local cover $\pi_{S,A}\colon (S_A, \widehat{A}) \to (S, A)$; every morphism g has a unique commuting lift $\widehat{g}$, again a morphism, and these assignments are functorial with π natural [Kah23, Theorem 7.12, pp. 44–45].

2.2. What is used from the sources.

Proposition 2.3 (Facts from the sources)**.** (i) *Let (S, A) be a marked Riemann surface with $S \setminus A$ connected and let U be a local neighborhood of A. Then $S_A \setminus \widehat{A}$ is the connected cover of $S \setminus A$ classified by the image of $\pi_1(U \setminus A)$ in $\pi_1(S \setminus A)$, and S_A is obtained from it*

by regluing U along its canonical lift [Kah23, Theorem 7.12 and its proof, p. 44].

(ii) *Localization preserves immersions, and it carries a morphism restricting to a covering map off the marked sets to a branched covering; in the dynamical situation of* (iii) *the localized dynamical leg is a branched covering of degree equal to the local degree $d_{A'}$ of f at the marked preimage A'* [Kah23, §§7.4.1–7.4.2, p. 45].

(iii) *Dynamical localization of a ψ-polynomial-like $(i, f)\colon \mathbf{U}' \to \mathbf{U}$ at an FJ-set $A \subset K$, with $A' = i^{-1}(A)$ and with removal of a compact $B \subset K$ satisfying the hypotheses of the source (A is a component of B, $\mathbf{U} \setminus B$ is connected, A' is a component of $f^{-1}(A)$, $i^{-1}(B) \subset f^{-1}(B)$, and all branch values of f lie in B), is performed on the marked surfaces*

$$T = (\mathbf{U} \setminus B) \cup A, \qquad T' = (\mathbf{U}' \setminus f^{-1}B) \cup A'$$

[Kah23, §7.4.2, p. 45]*; the resulting pair $(\widehat{\imath}, \widehat{f})\colon (T'_{A'}, \widehat{A'}) \to (T_A, \widehat{A})$ is again a ψ-polynomial-like map of degree $d_{A'}$, with $\widehat{\imath}$ an immersion, $\widehat{f}$ a proper branched covering, and $\widehat{\imath}^{-1}(\widehat{A}) = \widehat{f}^{-1}(\widehat{A}) = \widehat{A'}$.*

(iv) *The* canonical renormalization *of period p of a ψ-quadratic-like map that is primitively renormalizable of period p is the construction of* (iii) *applied to the iterate (i^p, f^p) with $A = K_p$ and $B = \mathcal{K}_p$* [Kah23, §7.4.3, pp. 45–46]*; the central local degree is* 2 [Kah23, §7.4.2, p. 45, with §1.1].

(v) *At the quadratic-like stage the map i^p is the inclusion of the nested domains $U^p \subset U^0$, and every restriction $f\colon U^{r+1} \to U^r$ of $f\colon U^1 \to U^0$ is again quadratic-like* [Kah23, §6, p. 31 and footnote 15].

(vi) *The pullback tower over a ψ-polynomial-like map is realized in equalizer coordinates: with $(i_r, f_r)\colon (\mathbf{U}^{r+1}, K^{r+1}) \to (\mathbf{U}^r, K^r)$,*

$$\mathbf{U}^{r+2} = \{(x, y) \in \mathbf{U}^{r+1} \times \mathbf{U}^{r+1} : \, i_r(x) = f_r(y)\}, \qquad i_{r+1}(x, y) = y, \quad f_{r+1}(x, y) = x,$$

each stage is again ψ-polynomial-like of the same degree, the squares $i_r \circ f_{r+1} = f_r \circ i_{r+1}$ commute, and each square is a categorical pullback of the previous stage [Kah23, §§7.3.1–7.3.2, pp. 38–43].

(vii) *Every ψ-polynomial-like map of degree $d \geq 2$ admits a restriction on which the immersion leg is an embedding: there is an open set $W \subset \mathbf{U}$ containing K such that, with $W' = f^{-1}(W)$, the map $f\colon W' \to W$ is a proper degree-d branched covering, $i|_{W'}$ is an embedding, $i(W') \subset W$, and $\operatorname{mod}\big(W \setminus i(W')\big) \geq \mu > 0$ with μ uniform for d bounded and $\operatorname{mod}(\mathbf{U} \setminus K)$ bounded below* [Kah23, Theorem 7.3, p. 37]*; the statement is restated in* [KL08, Lemma 3.1, p. 10].

Proof. These are the cited statements. □

2.3. **The ordinary primitive canonical seed.**

Definition 2.4 (Ordinary primitive canonical seed)**.** Let $f\colon U^1 \to U^0$ be an ordinary quadratic-like restriction of a quadratic polynomial, primitively

renormalizable of period $p > 1$, and set $U^r = f^{-r}(U^0)$, so that $U^{r+1} \subset U^r$ and every restriction $f\colon U^{r+1} \to U^r$ is quadratic-like (Proposition 2.3(v)). Write $F = f^p\colon U^p \to U^0$, a proper holomorphic map of degree 2^p, and let $i^p\colon U^p \hookrightarrow U^0$ be the physical inclusion. Let K_p be the filled Julia set of the p-renormalization, put $K_p(j) = f^j(K_p)$ for $0 \le j < p$ and $\mathcal{K}_p = \bigcup_{j=0}^{p-1} K_p(j)$; primitivity means that the $K_p(j)$ are pairwise disjoint [Kah23, §1.1]. Set $A = K_p$ and $B = \mathcal{K}_p$. The marked surfaces of the canonical localization of Proposition 2.3(iv) are then[1]

$$T' = (U^p \setminus F^{-1}B) \cup A, \qquad T = (U^0 \setminus B) \cup A, \tag{2.1}$$

with complementary surfaces

$$\Omega_s = T' \setminus A = U^p \setminus F^{-1}B, \qquad \Omega_t = T \setminus A = U^0 \setminus B,$$

and $a = i^p\colon T' \to T$ is the restriction of the physical inclusion.[2] Since $F(K_p(j)) = K_p(j)$ for every j, we have $B \cap U^p \subset F^{-1}B$ and therefore $\Omega_s \subset \Omega_t$. Together with the based data of Definition 2.6 below we call this configuration an *ordinary primitive canonical seed*; it is depicted in Figure 1.

2.4. **Jordan local neighborhoods and shared based data.**

Lemma 2.5 (Jordan neighborhoods are local pairs)**.** *Let $A \subset \mathbb{C}$ be a non-degenerate full continuum compactly contained in a Jordan disk D_A. Then $D_A \setminus A$ is a topological annulus; in particular (D_A, A) is a local pair in the sense of Definition 2.2.*

Proof. A connected component of $D_A \setminus A$ whose closure does not meet ∂D_A would be a bounded component of $\mathbb{C} \setminus A$, contradicting fullness of A; hence $D_A \setminus A$ is connected. In the sphere $\widehat{\mathbb{C}}$ the complement of $D_A \setminus A$ has exactly the two components A and $\widehat{\mathbb{C}} \setminus D_A$, so $D_A \setminus A$ is a doubly connected plane domain, that is, an annulus. Its fundamental group is infinite cyclic, generated by a core curve, which is the defining property of a local pair. □

[1] This choice of the removal set B is not the only one available in the source. Kahn describes at [Kah23, §7.4.4, p. 46] two further natural renormalizations, obtained as dynamical localizations of the *same* iterate (i^p, f^p) at the *same* marked continuum $A = K_p$ but with a different removal set: one takes $B := A \cup \mathcal{K}_{pq}$, and a second is built from the postcritical set and the geodesic γ_p. Throughout this paper "canonical renormalization" means the construction of [Kah23, §7.4.3, pp. 45–46] with $B = \mathcal{K}_p$, and the restriction is essential rather than decorative: the argument of Section 4 turns on the source removing all of $F^{-1}B$ while the target removes only B, and enlarging B to $A \cup \mathcal{K}_{pq}$ changes that asymmetry. We make no claim about the variants of [Kah23, §7.4.4, p. 46].

[2] The hypothesis list for dynamical localization at [Kah23, §7.4.2, p. 45] reads $A' = i^{-1}(A)$, and [Kah23, §7.4.3, p. 45] then writes $K'_p = i^{-1}(K_p)$ while simultaneously instructing the reader to replace (i, f) by the iterate $(i^p, f^p)\colon U^p \to U^0$. We use throughout the intended reading, in which the symbol i denotes the immersion leg of the map actually being localized: at the ordinary primitive seed $A' = (i^p)^{-1}(A) = A$, since $K_p \subset U^p$, so the marked continuum is literally the same set on both sides.

Kahn's own requirement on a local neighborhood is weaker than the one adopted here: he asks only that every component of $U \setminus A$ be an annulus [Kah23, §7.4.1, p. 44]. Restricting to *Jordan* neighborhoods is a specialization, legitimate by Lemma 2.5, and convenient because it makes the winding computations of Section 4 immediate.

Definition 2.6 (Shared local neighborhood and based representatives)**.** Fix once and for all a Jordan disk $D_A \supset A$ with $\overline{D_A} \subset U^p$ and $\overline{D_A} \cap (F^{-1}B \setminus A) = \varnothing$. Such a disk exists: by Corollary 3.5 the set $F^{-1}B$ is a finite pairwise-disjoint family of full continua, and A is one of them, since A is a component of $F^{-1}(A)$ by Proposition 2.3(iii)–(iv) while the preimages of the pairwise disjoint sets $K_p(j)$ are disjoint; now apply Corollary 3.6 to that family. Since $D_A \setminus A \subset \Omega_s \subset \Omega_t$, the same disk is a valid local neighborhood on both sides, and since a is the identity on $D_A \setminus A$ a single basepoint $z_* \in D_A \setminus A$ and a single choice of connecting paths serve for both. Set

$$H_s = \operatorname{im}\big[\pi_1(D_A \setminus A, z_*) \to \pi_1(\Omega_s, z_*)\big], \qquad H_t = \operatorname{im}\big[\pi_1(D_A \setminus A, z_*) \to \pi_1(\Omega_t, z_*)\big]. \tag{2.2}$$

These are the based representatives of the localization subgroups used throughout. By Proposition 2.3(i) the associated connected based covers $\pi_s \colon \widehat{\Omega}_s \to \Omega_s$ and $\pi_t \colon \widehat{\Omega}_t \to \Omega_t$ are the localization covers of the two complementary surfaces; we write $\widehat{a} \colon \widehat{\Omega}_s \to \widehat{\Omega}_t$ for the unique commuting lift of a, and $\widehat{\imath} \colon T'_A \to T_A$ for the localized inclusion between the completed localized marked surfaces.

The unbased localization subgroups are well defined only up to conjugacy, and the choice of based representative in (2.2) is immaterial for what follows: the witnessing data produced in Section 4 is detected by a winding homomorphism that annihilates every conjugate of H_s.

Lemma 2.7 (Injectivity of the completion)**.** *In the connected-complement localization construction of Proposition 2.3(i), the natural map from the complement covering surface $\widehat{\Omega}$ into the completed localized surface S_A is injective.*

Proof. The completed surface is the quotient of the disjoint union $U \sqcup \widehat{\Omega}$ in which each point $u \in U \setminus A$ is identified with its image $\ell(u) \in \widehat{\Omega}$ under the canonical lift ℓ of $U \setminus A$. Every identification therefore pairs one point of U with one point of $\widehat{\Omega}$, and no two points of $\widehat{\Omega}$ are ever identified with one another, since ℓ is single-valued. Hence the composition $\widehat{\Omega} \hookrightarrow U \sqcup \widehat{\Omega} \to S_A$ is injective. □

Applied on both sides of the canonical seed, Lemma 2.7 shows that the filled completion identifies no pair of distinct complement-cover points. Consequently every point collision of the completed localized inclusion $\widehat{\imath}$ arises from a collision of the complement-cover lift $\widehat{a}$, and conversely; this is the bridge used in Theorem 4.5.

3. Subgroup reflection and peripheral isolation

Two independent tools are proved here. The first converts injectivity of a lift between two based covers into an exact equality of subgroups and, when the equality fails, converts each offending class into an explicit pair of colliding points. The second is a package of plane-topological statements about full continua and their pullbacks under quadratic-like maps, supplying the isolating Jordan disks and the connectivity used to build peripheral loops in the source complement.

3.1. **Local conventions.** A *full continuum* is a nonempty compact connected set $K \subset \mathbb{C}$ whose complement $\mathbb{C} \setminus K$ is connected, and $K \Subset U$ means that K is compactly contained in U. Every space to which covering-space theory is applied below is connected, locally path-connected and semilocally simply connected; for a subgroup $H \leq \pi_1(X, x)$ we write $X_H \to X$ for the connected based cover classified by H. The lifting criterion and the endpoint criterion for covering spaces are used in the standard form of [Hat02, §1.3]. The seed data F, A, B, Ω_s, Ω_t, D_A, z_*, H_s, H_t are those of Definitions 2.4 and 2.6 and are not redefined here: the results below are stated for abstract covering data and for arbitrary full continua, and are specialized to the seed in Sections 4 and 6.

3.2. **The subgroup-reflection criterion.**

Theorem 3.1 (Subgroup-reflection criterion)**.** *Let X_s, X_t be connected, locally path-connected, semilocally simply connected spaces, let $p_s\colon \widehat{X}_s \to X_s$ and $p_t\colon \widehat{X}_t \to X_t$ be connected covers classified by subgroups $H_s \leq \pi_1(X_s, x_s)$ and $H_t \leq \pi_1(X_t, x_t)$, with base lifts $\widehat{x}_s \in p_s^{-1}(x_s)$ and $\widehat{x}_t \in p_t^{-1}(x_t)$, and let $a\colon (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based embedding. Put*

$$K_{\mathrm{ref}} = a_*^{-1}(H_t) \;\leq\; \pi_1(X_s, x_s), \tag{3.1}$$

the reflected subgroup*. Then:*

1. *a based lift $\widehat{a}\colon \widehat{X}_s \to \widehat{X}_t$, that is a map with $p_t \circ \widehat{a} = a \circ p_s$ and $\widehat{a}(\widehat{x}_s) = \widehat{x}_t$, exists if and only if $a_*(H_s) \subseteq H_t$, equivalently if and only if $H_s \leq K_{\mathrm{ref}}$, and it is then unique;*
2. *such a lift $\widehat{a}$ is injective if and only if $K_{\mathrm{ref}} = H_s$;*
3. *if that equality fails, then every class $[\gamma] \in K_{\mathrm{ref}} \setminus H_s$ is witnessing data for an explicit point collision: the lift of γ to $\widehat{X}_s$ starting at the base lift has distinct endpoints, and $\widehat{a}$ identifies those two endpoints.*

Proof. The lifting criterion states that a based lift $\widehat{a}$ exists, and is then unique, if and only if

$$a_*\big(p_{s*}\pi_1(\widehat{X}_s, \widehat{x}_s)\big) \subseteq p_{t*}\pi_1(\widehat{X}_t, \widehat{x}_t),$$

which is exactly $a_*(H_s) \subseteq H_t$, and this in turn is exactly $H_s \subseteq a_*^{-1}(H_t) = K_{\mathrm{ref}}$. This proves (1), and in particular the inclusion $H_s \leq K_{\mathrm{ref}}$ holds whenever $\widehat{a}$ exists.

Step 1: the reflection equality implies injectivity. Assume $K_{\mathrm{ref}} = H_s$ and let $\widehat{u}, \widehat{v} \in \widehat{X}_s$ satisfy $\widehat{a}(\widehat{u}) = \widehat{a}(\widehat{v})$. Projecting and using $p_t \circ \widehat{a} = a \circ p_s$,

$$a\big(p_s(\widehat{u})\big) = p_t\big(\widehat{a}(\widehat{u})\big) = p_t\big(\widehat{a}(\widehat{v})\big) = a\big(p_s(\widehat{v})\big),$$

and a is injective, so $p_s(\widehat{u}) = p_s(\widehat{v}) =: u$. Choose paths $\widehat{\alpha}, \widehat{\beta}$ in $\widehat{X}_s$ from $\widehat{x}_s$ to $\widehat{u}$ and to $\widehat{v}$ respectively, and set $\alpha = p_s \circ \widehat{\alpha}$ and $\beta = p_s \circ \widehat{\beta}$; both run from x_s to u. By the endpoint criterion the two source endpoints agree, that is $\widehat{u} = \widehat{v}$, if and only if the loop $\gamma = \alpha * \beta^{-1}$ has class in H_s.

On the target side, $\widehat{a} \circ \widehat{\alpha}$ and $\widehat{a} \circ \widehat{\beta}$ are the lifts, from the common base lift $\widehat{x}_t$, of the paths $a \circ \alpha$ and $a \circ \beta$. The hypothesis $\widehat{a}(\widehat{u}) = \widehat{a}(\widehat{v})$ says that these two lifts have the same endpoint, so the endpoint criterion applied in the target gives $a_*[\gamma] = \big[(a \circ \alpha) * (a \circ \beta)^{-1}\big] \in H_t$, that is $[\gamma] \in K_{\mathrm{ref}}$. The assumed equality gives $[\gamma] \in H_s$, hence $\widehat{u} = \widehat{v}$. Thus $\widehat{a}$ is injective.

Step 2: injectivity implies the reflection equality. Assume $\widehat{a}$ is injective. The inclusion $H_s \subseteq K_{\mathrm{ref}}$ is the lift-existence statement already proved. For the reverse inclusion let $[\gamma] \in K_{\mathrm{ref}}$, lift γ to $\widehat{X}_s$ from $\widehat{x}_s$, and call the endpoint $\widehat{y}_s$. The path $\widehat{a} \circ \widehat{\gamma}$ is the lift of $a \circ \gamma$ from $\widehat{x}_t$; since $a_*[\gamma] \in H_t$, that target lift is closed. Consequently $\widehat{a}(\widehat{y}_s) = \widehat{a}(\widehat{x}_s)$, and injectivity of $\widehat{a}$ gives $\widehat{y}_s = \widehat{x}_s$, that is, the source lift of γ is closed and $[\gamma] \in H_s$. Hence $K_{\mathrm{ref}} \subseteq H_s$, and with the previous inclusion this proves the equivalence (2).

Step 3: explicit failure data. Let $[\gamma] \in K_{\mathrm{ref}} \setminus H_s$. Then the source lift of γ begins at $\widehat{x}_s$ and ends at a point $\widehat{y}_s$ distinct from $\widehat{x}_s$, because $[\gamma] \notin H_s$, while the target lift of $a \circ \gamma$ is closed, because $a_*[\gamma] \in H_t$. Therefore

$$\widehat{y}_s \neq \widehat{x}_s, \qquad \widehat{a}(\widehat{y}_s) = \widehat{a}(\widehat{x}_s).$$

This is a point-level collision of $\widehat{a}$, not merely the failure of a combinatorial labelling: the class $[\gamma]$ records the exact sheet displacement that becomes invisible in the target cover. □

Remark 3.2 (Path formulation). Call two paths α, β in X_s with common endpoints *source equivalent* if $[\alpha * \beta^{-1}] \in H_s$, and *target equivalent* if $a_*[\alpha * \beta^{-1}] \in H_t$. The endpoint criterion detects each relation on the corresponding cover, so Theorem 3.1 says precisely that $\widehat{a}$ is injective if and only if target equivalence of source paths implies source equivalence; the opposite implication is automatic from $a_*(H_s) \subseteq H_t$. In Kahn's path model of the canonical renormalization [Kah23, §7.4.5, pp. 46–47] it is the automatic direction that is proved: source-equivalent paths are carried to target-equivalent paths.

3.3. **Peripheral isolation.**

Lemma 3.3 (Jordan-neighborhood basis)**.** *Let $K \Subset U \subset \mathbb{C}$ be a full continuum and let $\mathcal{O}$ be a neighborhood of K compactly contained in U. Then there is a Jordan domain V with*

$$K \subset V \subset \overline{V} \subset \mathcal{O}. \tag{3.2}$$

Moreover there is a nested sequence of such Jordan domains with

$$K \subset V_{r+1}, \qquad \overline{V}_{r+1} \subset V_r, \qquad \bigcap_{r\geq 1} \overline{V}_r = K. \tag{3.3}$$

Proof. If K is a singleton, round disks centered at that point satisfy (3.2) and (3.3). Otherwise fullness and connectedness make $\widehat{\mathbb{C}} \setminus K$ a simply connected domain of the sphere containing ∞. Uniformize it by the exterior of the closed unit disk, normalized at infinity: let $\Phi\colon \{|w| > 1\} \cup \{\infty\} \to \widehat{\mathbb{C}} \setminus K$ be the conformal isomorphism with $\Phi(\infty) = \infty$. For each radius $\rho > 1$ the image $\Phi(\{|w| = \rho\})$ is a Jordan curve in the complement of K, and its bounded complementary side $V(\rho)$ contains K. Every point z of the complement of K has uniformizing modulus $|\Phi^{-1}(z)|$ strictly greater than 1, so $z \notin \overline{V(\rho)}$ as soon as $\rho < |\Phi^{-1}(z)|$; hence $\bigcap_{\rho>1} \overline{V(\rho)} = K$. The sets $\overline{V(\rho)} \setminus \mathcal{O}$ are compact, decrease as $\rho \downarrow 1$, and have empty intersection, so $V = V(\rho)$ satisfies $\overline{V} \subset \mathcal{O}$ once ρ is close enough to 1; this is (3.2). Setting $V_r = V(\rho_r)$ for a strictly decreasing sequence $\rho_r \downarrow 1$ gives the nested sequence of (3.3), whose intersection is K because every point of the complement has uniformizing modulus strictly greater than 1. □

Lemma 3.4 (One quadratic pullback)**.** *Let $q\colon U' \to U$ be a proper holomorphic map of degree two between plane topological disks, let $K \Subset U$ be a full continuum, let c be the unique critical point of q, and let $v = q(c)$.*

1. *If $v \in K$, then $q^{-1}K$ is a single full continuum, and the local degree of q on it is two.*
2. *If $v \notin K$, then $q^{-1}K$ is the disjoint union of two full continua, each mapped homeomorphically onto K by q.*

Proof. Choose the nested Jordan basis V_r of Lemma 3.3, with each ∂V_r avoiding v; if $v \notin K$, choose in addition $\overline{V}_1$ disjoint from v.

Case $v \notin K$. Then $v \notin V_r$ for every r, so $q\colon q^{-1}(V_r) \to V_r$ is an unbranched proper covering of total degree two. Since V_r is simply connected, its preimage has exactly two components W_r^0, W_r^1 and each restriction $q\colon W_r^k \to V_r$ is a biholomorphism. The two inverse branches of q on V_1 label the components coherently for all r, and the nesting $\overline{V}_{r+1} \subset V_r$ gives $\overline{W}_{r+1}^k \subset W_r^k$ for $k = 0, 1$. Set $L_k = \bigcap_{r\geq 1} \overline{W}_r^k$. As a nested intersection of nonempty compact connected sets, each L_k is nonempty, compact and connected. It is full, because

$$\mathbb{C} \setminus L_k = \bigcup_{r\geq 1} \big(\mathbb{C} \setminus \overline{W}_r^k\big)$$

is an increasing union of connected sets: each $\overline{W}_r^k$ is a closed Jordan disk, since ∂V_r avoids the critical value and the inverse branch carries V_r and its regular boundary biholomorphically. The inverse branches identify each L_k homeomorphically with $K = \bigcap_r \overline{V}_r$, and $q^{-1}K = L_0 \sqcup L_1$.

Case $v \in K$. Every V_r contains v. The component W_r of $q^{-1}(V_r)$ containing the critical point c maps with degree at least two; since the total degree is two, its degree is exactly two and W_r exhausts $q^{-1}(V_r)$. Riemann–Hurwitz for the proper degree-two map $W_r \to V_r$ with its single simple critical point gives $\chi(W_r) = 2\chi(V_r) - 1 = 1$, so W_r is a disk; and ∂W_r is a regular Jordan curve because ∂V_r contains no critical value. The compact Jordan disks $\overline{W}_r$ are nested, with $q^{-1}K = \bigcap_{r\geq 1} \overline{W}_r$, and the same increasing-complement argument as in the previous case shows that this intersection is connected and full.

The bordered Riemann–Hurwitz step used here can be obtained by doubling W_r and V_r across their regular boundaries: the doubled degree-two map of closed surfaces has exactly the critical point and its mirror, so $2 - 2g = 2 \cdot 2 - 2 = 2$, whence $g = 0$; since the double of a planar domain with b boundary components has genus $b - 1$, we get $b = 1$, that is, W_r is a disk with Jordan boundary. □

Corollary 3.5 (Finite full p-fold pullback)**.** *Under a chain of p quadratic-like pullbacks, the inverse image of one full continuum is a union of at most 2^p pairwise disjoint full continua, and every resulting component maps onto its parent continuum with positive integer degree. The inverse image of a finite pairwise-disjoint family of full continua is again a finite pairwise-disjoint family of full continua.*

Proof. Apply Lemma 3.4 successively along the chain. At each stage every parent continuum has either one degree-two child or two degree-one children, and every child maps onto its parent. Children of different parents are disjoint, because their images lie in the disjoint parent continua. Induction over the p stages gives the bound 2^p, the stated surjectivity, and a positive product degree for every terminal component. Applying this separately to the members of a finite pairwise-disjoint family proves the last statement. □

Corollary 3.6 (Isolating Jordan disks)**.** *Let $C_1, \dots, C_N \Subset U$ be pairwise disjoint full continua. For each i there is a Jordan domain $D_i \Subset U$ with*

$$C_i \subset D_i, \qquad \overline{D}_i \cap C_j = \varnothing \quad (j \neq i), \tag{3.4}$$

and the D_i may be chosen pairwise disjoint.

Proof. A finite pairwise-disjoint family of compacta in U has positive minimum pairwise distance and positive distance from ∂U. Apply Lemma 3.3 inside pairwise-disjoint metric neighborhoods of the C_i whose closures lie in U. □

Lemma 3.7 (Connectivity after finitely many full deletions)**.** *Let U be a plane topological disk and let $C_1, \dots, C_N \Subset U$ be pairwise disjoint full continua. Then $U \setminus \bigcup_{i=1}^{N} C_i$ is path connected.*

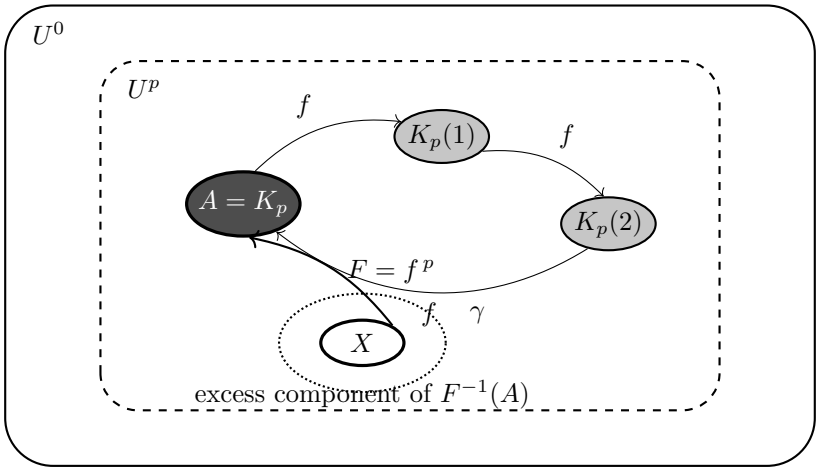


FIGURE 1. The canonical seed of a primitive renormalization of period p (drawn for $p = 3$). The cycle $B = \mathcal{K}_p = \bigcup_j K_p(j)$ is removed and the central continuum $A = K_p$ is retained on both sides. Because $\deg F = 2^p$ exceeds the central degree 2, the preimage $F^{-1}(A)$ has a component $X \neq A$. A based loop γ encircling X lies outside the source localization subgroup but is contracted by the physical inclusion in the target: its localized lift has distinct endpoints that the localized inclusion identifies.

Proof. Applying Lemma 3.3 twice inside the disjoint neighborhoods used in Corollary 3.6, choose pairwise-disjoint Jordan domains

$$C_i \subset D_i^-, \qquad \overline{D_i^-} \subset D_i^+, \qquad \overline{D_i^+} \subset U.$$

The finitely connected plane domain $U_0 = U \setminus \bigcup_{i=1}^N \overline{D_i^-}$ is connected: after a homeomorphism $U \cong \mathbb{D}$, removing finitely many pairwise-disjoint closed Jordan disks compactly contained in $\mathbb{D}$ leaves a disk with finitely many holes. As an open connected subset of the plane, U_0 is path connected.

For each i the open set $D_i^+ \setminus C_i$ is connected. Indeed $\mathbb{C} \setminus C_i$ is connected by fullness, and its intersection with the Jordan disk D_i^+ cannot have a component separated from the outer collar of ∂D_i^+: such a component would be a bounded complementary component of C_i, contradicting fullness. Hence $D_i^+ \setminus C_i$ is path connected. It contains the nonempty annulus $D_i^+ \setminus \overline{D_i^-}$, which is also contained in U_0; thus every $D_i^+ \setminus C_i$ meets U_0 in a nonempty connected set. The union of U_0 with these sets is therefore path connected, and

$$U_0 \cup \bigcup_{i=1}^N (D_i^+ \setminus C_i) = U \setminus \bigcup_{i=1}^N C_i. \qquad \square$$

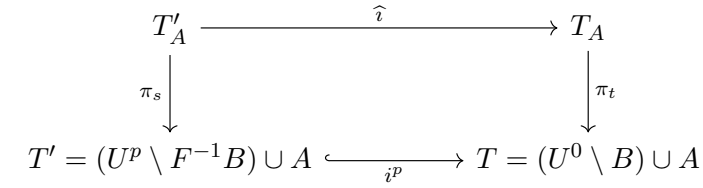


FIGURE 2. The localization covers. The covers π_s, π_t are the based covers classified by the localization subgroups $H_s \leq \pi_1(T' \setminus A)$ and $H_t \leq \pi_1(T \setminus A)$; the localized inclusion $\widehat{\imath}$ is the unique based lift of the physical inclusion i^p. The forced collision is the statement that $\widehat{\imath}$ is never injective at an ordinary primitive canonical seed of period $p > 1$: the class $[\gamma]$ of Figure 1 lies outside H_s while $(i^p)_*[\gamma]$ is trivial in H_t.

4. Forced noninjectivity at the canonical seed

This section proves Theorem A, and the mechanism is summarized by Figures 1 and 2: the degree budget ($\deg F = 2^p$ against central degree 2) produces an excess component X of $F^{-1}(A)$; peripheral isolation puts an isolating Jordan disk around X; the winding number about a point of X kills the source localization subgroup but not the peripheral class of that disk; and the target contracts the peripheral loop across the disk, because the target removes only B while the source removes all of $F^{-1}B$. Throughout, $F = f^p \colon U^p \to U^0$, $A = K_p$, $B = \mathcal{K}_p$, $\Omega_s = U^p \setminus F^{-1}B$, $\Omega_t = U^0 \setminus B$, and the basepoint z_*, the shared Jordan disk $D_A \supset A$ and the based localization subgroups H_s, H_t are those fixed in Definition 2.6.

Proposition 4.1 (Excess central pullback)**.** *At every ordinary primitive canonical seed of period $p > 1$:*

(1) *$B \cap U^p \subset F^{-1}B$; consequently $\Omega_s \subset \Omega_t$.*
(2) *There is a component*
$$X \subset F^{-1}A, \qquad X \neq A. \tag{4.1}$$
(3) *Every such component satisfies*
$$X \cap B = \varnothing. \tag{4.2}$$
(4) *Every such component admits a Jordan domain $D_X \Subset U^p$ with*
$$X \subset D_X, \qquad \overline{D_X} \cap \big(F^{-1}B \setminus X\big) = \varnothing, \qquad \overline{D_X} \cap \overline{D_A} = \varnothing; \tag{4.3}$$
in particular $\partial D_X \subset \Omega_s$ and $\overline{D_X} \subset \Omega_t$.

Proof. For $0 \leq j < p$ write $K_j = f^j(A)$, so that $K_0 = A$ and $B = \bigcup_j K_j$. Since $F = f^p$ commutes with f^j on these sets and $F(A) = A$,
$$F(K_j) = f^j\big(F(A)\big) = f^j(A) = K_j. \tag{4.4}$$

Hence $K_j \cap U^p \subset F^{-1}K_j \subset F^{-1}B$ for every j, which is $B \cap U^p \subset F^{-1}B$. (Only the relative form is available, and only it is needed: $F^{-1}B$ lies in U^p, whereas the noncentral cycle members K_j, $j \neq 0$, need not.) Therefore, if $x \in \Omega_s = U^p \setminus F^{-1}B$ lay in B, it would lie in $B \cap U^p \subset F^{-1}B$, a contradiction; so $\Omega_s \subset U^0 \setminus B = \Omega_t$, proving (1).

For (2), the map F is proper of degree 2^p [Kah23, §6, p. 31], while by the definition of the primitive quadratic-like renormalization the central component A has local degree two under F [Kah23, §7.4.2, p. 45, with §1.1]; both facts are recorded in Section 2. Using Lemma 3.3, choose a Jordan domain $V \supset A$ compactly contained in the target domain of the central degree-two polynomial-like restriction of F at A. The component V_0 of $F^{-1}(V)$ containing A then maps onto V with degree two. By properness every component of $F^{-1}(V)$ maps onto V with a positive integer degree and these degrees sum to $\deg F = 2^p$; since $p > 1$ we have $2^p - 2 > 0$, so there is a component $W \neq V_0$. As $F\colon W \to V$ is onto, W contains a point of $F^{-1}(A)$; the component X of $F^{-1}(A)$ through that point is connected and contained in $F^{-1}(V)$, hence contained in the single component W. Since $A \subset V_0$ and $W \cap V_0 = \varnothing$, we get $X \neq A$, which is (4.1).

For (3), suppose $X \cap K_j \neq \varnothing$. If $j \neq 0$, a point of the intersection is carried by F into A (because $X \subset F^{-1}A$) and also into K_j (by (4.4)), contradicting the primitive disjointness $A \cap K_j = \varnothing$. If $j = 0$ then $X \cap A \neq \varnothing$, so the connected sets X and A lie in one component of $F^{-1}A$, forcing $X = A$. This is (4.2).

For (4), F is a composition of p proper degree-two maps between plane topological disks and B is a finite disjoint family of full continua, so by Corollary 3.5 the set $F^{-1}B$ is again a finite pairwise-disjoint family of full continua compactly contained in U^p, to which every component of $F^{-1}A$, in particular X, belongs. Definition 2.6 gives $\overline{D_A} \cap X = \varnothing$. Thus the compact sets

$$X \quad \text{and} \quad \overline{D_A} \cup \left(F^{-1}B \setminus X\right)$$

are disjoint. Choose a neighborhood of X compactly contained in U^p whose closure misses the second set, and apply Lemma 3.3 inside that neighborhood. The resulting Jordan domain D_X satisfies (4.3), including the stated separation from $\overline{D_A}$. Since its boundary misses both X and $F^{-1}B \setminus X$, we have $\partial D_X \subset \Omega_s$. Finally, since $\overline{D_X} \subset U^p$, any point of $\overline{D_X} \cap B$ would lie in $B \cap U^p \subset F^{-1}B$ by (1), and hence, by (4.2), in $F^{-1}B \setminus X$, contradicting (4.3). So $\overline{D_X} \cap B = \varnothing$, while $\overline{D_X} \subset U^p \subset U^0$; hence $\overline{D_X} \subset \Omega_t$. □

Definition 4.2 (Excess component)**.** An *excess component* of the canonical seed is a component $X \neq A$ of $F^{-1}(A)$. For the physical inclusion $A' = (i^p)^{-1}(A) = A$, so an excess component is a *different* component of $F^{-1}(A)$ from the marked one, and must not be confused with A'.

By Proposition 4.1(1) the physical inclusion restricts to a based embedding $a = i^p\colon \Omega_s \hookrightarrow \Omega_t$, and we write $K_{\mathrm{ref}} = a_*^{-1}(H_t) \leq \pi_1(\Omega_s, z_*)$ for the

reflected subgroup. Since a is the identity on $D_A \setminus A$, the based representatives of Definition 2.6 satisfy $a_*(H_s) = H_t$; in particular $H_s \le K_{\mathrm{ref}}$.

Remark 4.3 (Witnessing data for the excess component). By Corollary 3.5 the family of excess components is finite. Fix any excess component X; nothing below depends on the choice. Its successive images determine the component itinerary

$$\tau(X) = \big(X,\ f(X), \dots, f^p(X) = A;\ \varsigma_0, \dots, \varsigma_{p-1}\big), \tag{4.5}$$

where ς_r records the component of $f^{-1}\big(f^{r+1}X\big)$ containing $f^r X$. The access used below is the based homotopy class of ∂D_X in the complement of $F^{-1}B$.

Theorem 4.4 (Localization subgroup defect)**.** *Let X and D_X be as in Proposition 4.1, let $\eta \subset \Omega_s$ be any path from z_* to ∂D_X, and set*

$$\gamma = \eta * \partial D_X * \eta^{-1}, \tag{4.6}$$

with ∂D_X positively oriented. Then $[\gamma] \notin H_s$ while $a_*[\gamma] = 1 \in H_t$, and consequently

$$K_{\mathrm{ref}} = a_*^{-1}(H_t) \supsetneq H_s: \tag{4.7}$$

the exact failure of the subgroup-reflection equality of Theorem 3.1 on the canonical-localization data.

Proof. By Corollary 3.5 the removed family $F^{-1}B$ is a finite union of pairwise disjoint full continua, so $\Omega_s = U^p \backslash F^{-1}B$ is path connected by Lemma 3.7; hence η exists and γ in (4.6) is a loop based at z_*.

Fix $x \in X$. Since $x \in X \subset F^{-1}B$ is disjoint from Ω_s, winding number about x defines a homomorphism $\mathrm{Ind}_x \colon \pi_1(\Omega_s, z_*) \to \mathbb{Z}$. Every element of H_s is represented, before inclusion, by a loop in $D_A \setminus A$; by Definition 2.6 the disk D_A satisfies $\overline{D_A} \cap (F^{-1}B \setminus A) = \varnothing$, so $x \notin D_A$, and any loop inside the simply connected D_A is null-homotopic in $\mathbb{C} \setminus \{x\}$. Consequently

$$\mathrm{Ind}_x(H_s) = \{0\}. \tag{4.8}$$

The path and the reverse path in (4.6) cancel in winding number, while the positively oriented ∂D_X has winding number one about $x \in X \subset D_X$. Thus

$$\mathrm{Ind}_x([\gamma]) = 1, \qquad \text{hence} \qquad [\gamma] \notin H_s. \tag{4.9}$$

For the target, Proposition 4.1(4) gives $\overline{D_X} \subset \Omega_t$, so ∂D_X contracts across the disk D_X inside Ω_t and γ is null-homotopic there:

$$a_*[\gamma] = 1 \in H_t. \tag{4.10}$$

Combining (4.9) and (4.10) gives $[\gamma] \in K_{\mathrm{ref}} \setminus H_s$, which with $H_s \le K_{\mathrm{ref}}$ is (4.7). □

Theorem 4.5 (Forced collision at ordinary primitive canonical seeds)**.** *Let $f \colon U^1 \to U^0$ be an ordinary quadratic-like restriction of a quadratic polynomial, primitively renormalizable of period $p > 1$, and let $\widehat{\imath} \colon T'_A \to T_A$ be the localized inclusion of the canonical renormalization, that is, the dynamical localization of (i^p, F) at $A = K_p$ with removal of $B = \mathcal{K}_p$. Then $\widehat{\imath}$ is* not

injective. *More precisely, there are a component $X \neq A$ of $F^{-1}A$, a based loop $\gamma \subset \Omega_s$, and the localization subgroups H_s, H_t with*

$$[\gamma] \notin H_s, \qquad (i^p)_*[\gamma] = 1 \in H_t, \tag{4.11}$$

such that the lift of γ to the source localization cover has distinct endpoints which $\widehat{\imath}$ identifies.

Proof. Take X, D_X, γ, H_s, H_t from Proposition 4.1 and Theorem 4.4; then (4.11) is (4.9) together with (4.10).

Let $\pi_s\colon \widehat{\Omega}_s \to \Omega_s$ and $\pi_t\colon \widehat{\Omega}_t \to \Omega_t$ be the localization covers, that is, the based connected covers classified by H_s and H_t [Kah23, §7.4.1, Theorem 7.12, p. 44], and let $\widehat{a}\colon \widehat{\Omega}_s \to \widehat{\Omega}_t$ be the unique based lift of $a = i^p$ with $\pi_t\widehat{a} = a\pi_s$, which exists because $a_*(H_s) \leq H_t$. Let $\widehat{z}_* \in \pi_s^{-1}(z_*)$ be the distinguished base lift, lift γ from $\widehat{z}_*$, and let $\widehat{z}'_*$ be the terminal endpoint of the lifted path. By the covering-subgroup endpoint criterion — a loop lifts closed at $\widehat{z}_*$ exactly when its class lies in H_s, as in the proof of Theorem 3.1 — together with (4.9),

$$\widehat{z}'_* \neq \widehat{z}_*. \tag{4.12}$$

The $\widehat{a}$-image of the lifted path is the target lift, from $\widehat{a}(\widehat{z}_*)$, of the loop $a \circ \gamma$. By (4.10) the class of $a \circ \gamma$ is trivial, in particular lies in H_t, so that target lift is closed and

$$\widehat{a}(\widehat{z}'_*) = \widehat{a}(\widehat{z}_*). \tag{4.13}$$

Both points lie in the complement cover $\widehat{\Omega}_s$. The completed localized surfaces are obtained by adjoining the marked continuum A along a local neighborhood, and by Lemma 2.7 that completion identifies no two distinct points of the complement cover, on either side. Hence (4.12) and (4.13) exhibit two distinct points of T'_A with equal $\widehat{\imath}$-image in T_A: an actual point collision of the localized inclusion.

In path language: if α is any source path from the marked base access to z_*, the classes $[\alpha]$ and $[\alpha * \gamma]$ are distinct in the source localization and equal after target localization. The forward path-equivalence statement of the source [Kah23, §7.4.5, pp. 46–47] — source-equivalent paths become target-equivalent — is respected; it is the converse, never asserted there, that fails. □

Remark 4.6 (Witnessing data for the collision). The proof produces explicit data: the excess component X with its isolating disk D_X and itinerary (4.5), the peripheral class $[\gamma]$, the subgroups H_s, H_t, the two distinguished source cosets H_s and $H_s[\gamma]$, the distinct source points $\widehat{z}_* \neq \widehat{z}'_*$, and their common image $\widehat{a}(\widehat{z}_*)$. No critical-value crossing and no numerical approximation is used to create the collision.

Remark 4.7 (Compatibility with the embedded inner collar). The sources state that i *is* an embedding near the filled set: [Kah23, Theorem 7.3, p. 37] produces $U \supset K$ with $U' = f^{-1}U$ such that $i|_{U'}$ is an embedding and $\operatorname{mod}\big(U \setminus i(U')\big) \geq \mu > 0$; [Kah23, Lemma 7.4(2), p. 37] gives the uniformized

form, an embedding on a round sub-annulus adjacent to the inner end; and [KL08, Lemma 3.1] repeats that i is an embedding in a neighborhood of K'. Read quickly this looks like a contradiction of Theorem 4.5. It is not, and it is worth locating the collision precisely.

The peripheral part of the witness is remote from the central continuum: Proposition 4.1 gives $\overline{D_X} \cap \overline{D_A} = \varnothing$. The based loop $\gamma = \eta * \partial D_X * \eta^{-1}$, however, also contains the access path from $z_* \in D_A \setminus A$ to ∂D_X. Thus it would be incorrect to describe the whole physical loop as lying outside an inner collar. The precise reconciliation is instead a statement about sheets.

Let $\widehat{L}$ be the component of $\pi_s^{-1}(D_A \setminus A)$ containing the distinguished base lift $\widehat{z}_*$. By the definition of H_s, every loop in $D_A \setminus A$ maps into H_s; hence $\pi_s|_{\widehat{L}} \colon \widehat{L} \to D_A \setminus A$ is the one-sheeted connected pullback cover and is a homeomorphism. In particular $\widehat{z}_*$ is the unique point of $\widehat{L}$ over z_*. The terminal point $\widehat{z}'_*$ of the lifted γ is a different point over the same z_* by (4.12), and therefore lies on a different component of $\pi_s^{-1}(D_A \setminus A)$. The filled completion attaches the marked continuum along the distinguished component $\widehat{L}$, and the local embedded collar may be chosen inside that sheet. The collision is therefore between the distinguished local sheet and a remote sheet, not between two points of one embedded collar.

Nothing here weakens the sources; the inner-collar embedding remains a genuine input, both to the moduli estimates of the renormalization theory and to the annular normal form used at the tile level. What the sources do not supply is any extension of that statement to a global one, and Appendix A shows why none follows from the stated hypotheses: it exhibits a degree-one annular immersion satisfying all of them that is nevertheless noninjective, and a complete abstract ψ-quadratic-like pair with a remote self-collision compatible with the embedded inner collar. Theorem 4.5 proves noninjectivity at every ordinary primitive canonical seed.

Proposition 4.8 (Scope of the forced collision)**.** *The following are proved without further hypotheses.*

(1) *Every first canonical localization of an ordinary primitively p-renormalizable quadratic polynomial, $p > 1$, has the collision of Theorem 4.5, with the data of Remark 4.6.*
(2) *That collision propagates through every pullback depth of the same seed.*

Proof. Item (1) restates Theorem 4.5 together with Remarks 4.3 and 4.6. Item (2) restates Corollary 7.4, proved in Section 7: the canonical factorization of the localized lift, and with it the cardinality of every fiber, is reproduced by exact pullback base change at each stage of the tower. □

Consequences at Kahn–Lyubich decoration stages require an additional stage-alignment hypothesis and are developed in a sequel.

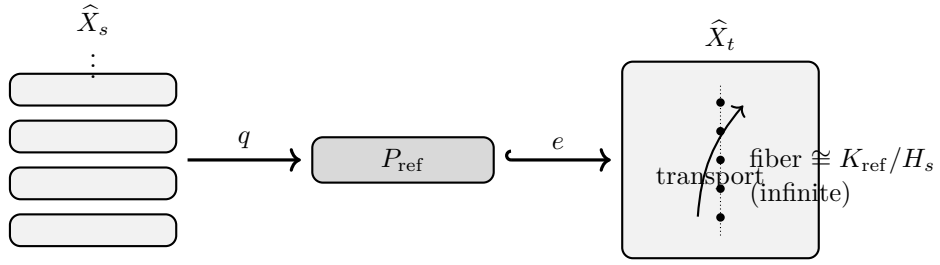


FIGURE 3. The canonical intermediate-cover resolution. The unresolved localized lift is replaced by the factorization $\widehat{X}_s \xrightarrow{q} P_{\mathrm{ref}} \xhookrightarrow{e} \widehat{X}_t$: a covering onto the embedded intermediate factor (unique in the based commuting comparison class) followed by an embedding. The sheet fiber over the embedded factor is a coset space detected by the winding homomorphism; it is infinite, and canonical transport moves sheets along paths in the base. Exact Kahn base change preserves the entire structure.

5. The canonical sheet resolution

Figure 3 shows the shape of the result: the localized lift of a based embedding factors as a covering onto an intermediate cover followed by a topological embedding, and the entire sheet defect of the lift is concentrated in the fiber of that covering.

5.1. **Standing data and conventions.** All spaces here are Hausdorff, connected, locally path-connected and semilocally simply connected, and all covers are connected and based; the planar complementary surfaces of the canonical localization satisfy these hypotheses. Path products and induced maps are based, and for a based covering $p\colon (\widehat{X},\widehat{x}) \to (X,x)$ we write $H = p_*\pi_1(\widehat{X},\widehat{x}) \le \pi_1(X,x)$ for its characteristic subgroup.

For subgroups $H \le K$ of any group (the case of interest below being $H_s \le K_{\mathrm{ref}}$),

$$K/H = \{\, kH : k \in K \,\} \tag{5.1}$$

denotes only the *pointed set* of left cosets, with distinguished point H; no multiplication on (5.1) is asserted unless $H \trianglelefteq K$. With the usual path-endpoint convention a cover fiber is naturally the right-coset set $H\backslash K$, and inversion $Hk \mapsto k^{-1}H$ is the canonical pointed bijection onto (5.1); the notation K/H therefore does not silently assume normality. Paths compose by the paper-wide rule: $\delta_1 * \delta_2$ traverses δ_1 first, and $[\delta_1][\delta_2] = [\delta_1 * \delta_2]$. Continuing the fiber point labelled H_sk by a based loop g appends g after k and gives H_skg; under the inversion identification this transport reads

$$k^{-1}H_s \longmapsto g^{-1}k^{-1}H_s, \tag{5.2}$$

which replaces any naive multiplication on K_{ref}/H_s and remains valid when H_s is not normal in K_{ref}.

Definition 5.1 (Canonical localization covering data)**.** Let $a\colon (X_s, x_s) \hookrightarrow (X_t, x_t)$ be a based topological embedding, and let $p_s\colon (\widehat{X}_s, \widehat{x}_s) \to (X_s, x_s)$ and $p_t\colon (\widehat{X}_t, \widehat{x}_t) \to (X_t, x_t)$ be connected based covers with characteristic subgroups H_s, H_t. Suppose a based lift $\widehat{a}\colon (\widehat{X}_s, \widehat{x}_s) \to (\widehat{X}_t, \widehat{x}_t)$ exists with

$$p_t\,\widehat{a} = a\,p_s. \tag{5.3}$$

The lifting criterion gives $a_*(H_s) \le H_t$; define the *reflected subgroup*

$$K_{\mathrm{ref}} := a_*^{-1}(H_t) \le \pi_1(X_s, x_s), \qquad \text{so that} \qquad H_s \le K_{\mathrm{ref}}. \tag{5.4}$$

Form the pullback

$$P = X_s \times_{X_t} \widehat{X}_t = \big\{(x, \widehat{y}) : a(x) = p_t(\widehat{y})\big\}, \tag{5.5}$$

let P_{ref} be the path component of the basepoint $(x_s, \widehat{x}_t)$, and write

$$r\colon P_{\mathrm{ref}} \to X_s, \quad r(x, \widehat{y}) = x, \qquad e\colon P_{\mathrm{ref}} \to \widehat{X}_t, \quad e(x, \widehat{y}) = \widehat{y}. \tag{5.6}$$

At the canonical seed of Section 4 this data is instantiated by $a = i^p$, $X_s = \Omega_s$, $X_t = \Omega_t$, with $\widehat{a}$ the complement-cover lift of Theorem 4.5; the completed localized inclusion $\widehat{\imath}$ is obtained from $\widehat{a}$ by the filled completion, which by Lemma 2.7 identifies no two complement-cover points.

5.2. **The resolution.**

Lemma 5.2 (Subgroup of the pullback component)**.** *With the data of Definition 5.1, the first projection $r\colon P_{\mathrm{ref}} \to X_s$ is the connected based cover of X_s corresponding to the subgroup $K_{\mathrm{ref}} = a_*^{-1}(H_t)$.*

Proof. The pullback of a covering is a covering, and P is locally path-connected, so its path components are open. The selected component P_{ref} maps onto X_s: for $x \in X_s$ choose a path from x_s to x; its lift to the pullback from $(x_s, \widehat{x}_t)$ stays in P_{ref} and ends over x. Hence $r\colon P_{\mathrm{ref}} \to X_s$ is itself a connected covering.

It remains to compute its subgroup. Let γ be a based loop in X_s; its lift to P_{ref} from $(x_s, \widehat{x}_t)$ is

$$t \longmapsto \big(\gamma(t), \widehat{a\gamma}(t)\big), \tag{5.7}$$

where $\widehat{a\gamma}$ is the lift of $a\gamma$ to $\widehat{X}_t$ from $\widehat{x}_t$. The lift (5.7) is closed if and only if $\widehat{a\gamma}$ is closed, which by the cover–subgroup criterion holds if and only if $a_*[\gamma] \in H_t$, that is, by (5.4), if and only if $[\gamma] \in K_{\mathrm{ref}}$. Therefore $r_*\pi_1\big(P_{\mathrm{ref}}, (x_s, \widehat{x}_t)\big) = K_{\mathrm{ref}}$, which also identifies the selected path component as the based connected cover classified by K_{ref}. □

Lemma 5.3 (The resolved leg is embedded)**.** *The second projection*

$$e\colon P_{\mathrm{ref}} \hookrightarrow \widehat{X}_t \tag{5.8}$$

is a topological embedding.

Proof. Consider the second projection on the entire pullback P; its image is $p_t^{-1}\big(a(X_s)\big)$. Because a is an embedding, the formula

$$\widehat{y} \longmapsto \big(a^{-1}(p_t(\widehat{y})),\, \widehat{y}\big) \tag{5.9}$$

is a continuous inverse on that image, so the second projection identifies P homeomorphically with $p_t^{-1}\big(a(X_s)\big)$. Restricting this homeomorphism to P_{ref} gives (5.8). □

Remark 5.4. The embedding hypothesis on a is essential: if a has a base-space self-intersection, equality of second coordinates in (5.5) need not determine the first coordinate, and Lemma 5.3 can fail before any subgroup defect is considered; compare the abstract examples of Appendix A.

Theorem 5.5 (Canonical sheet resolution)**.** *With the data of Definition 5.1, define*

$$q\colon \widehat{X}_s \longrightarrow P_{\mathrm{ref}}, \qquad q(\widehat{u}) = \big(p_s(\widehat{u}),\, \widehat{a}(\widehat{u})\big). \tag{5.10}$$

Then:

1. *q is the unique based map over X_s from the H_s-cover to the K_{ref}-cover;*
2. *q is a surjective covering map;*
3. *the original lift factors exactly as*
$$\widehat{a} = e \circ q; \tag{5.11}$$
4. *for all $\widehat{u}, \widehat{v} \in \widehat{X}_s$,*
$$\widehat{a}(\widehat{u}) = \widehat{a}(\widehat{v}) \quad\Longleftrightarrow\quad q(\widehat{u}) = q(\widehat{v}); \tag{5.12}$$
5. *equivalently, the kernel pairs agree:*
$$\widehat{X}_s \times_{P_{\mathrm{ref}}} \widehat{X}_s = \widehat{X}_s \times_{\widehat{X}_t} \widehat{X}_s; \tag{5.13}$$
6. *the fiber of q over the based point is the pointed coset set*
$$\mathcal{S}_{\mathrm{loc}} = K_{\mathrm{ref}}/H_s; \tag{5.14}$$
7. *q is injective if and only if $K_{\mathrm{ref}} = H_s$; and*
8. *if $[K_{\mathrm{ref}} : H_s] < \infty$, then q has exactly $[K_{\mathrm{ref}} : H_s]$ sheets. No finiteness is asserted when the index is infinite.*

Proof. The formula (5.10) lands in the pullback (5.5) because (5.3) gives $a\big(p_s(\widehat{u})\big) = p_t\big(\widehat{a}(\widehat{u})\big)$, and it lands in P_{ref} because $\widehat{X}_s$ is path connected and q sends the basepoint to $(x_s, \widehat{x}_t)$. Moreover

$$r\, q = p_s. \tag{5.15}$$

Since $H_s \le K_{\mathrm{ref}}$, the lifting criterion for covers over the common base X_s gives a unique based map from the H_s-cover to the K_{ref}-cover; by (5.15) and Lemma 5.2, (5.10) is that map, proving (1).

We verify surjectivity directly rather than by cover classification. Let $z \in P_{\mathrm{ref}}$ and choose a path δ in P_{ref} from the based point to z. Lift $r\delta$ to $\widehat{X}_s$ from $\widehat{x}_s$, obtaining $\widehat{\delta}$. Both δ and $q\widehat{\delta}$ are lifts of $r\delta$ to P_{ref} from the same

initial point, so uniqueness of path lifting gives $q\big(\widehat{\delta}(1)\big) = \delta(1) = z$, and q is onto. Now choose a path-connected neighborhood $U \subset X_s$ evenly covered by both p_s and r. Each component of $p_s^{-1}(U)$ is mapped by q into one component of $r^{-1}(U)$; because both projections to U are homeomorphisms and $rq = p_s$, that restriction of q is a homeomorphism onto its target sheet, and surjectivity ensures every target sheet is met. Hence q is a covering map, proving (2).

The second coordinate in (5.10) is $\widehat{a}(\widehat{u})$, so (5.11) is immediate, proving (3). Since e is injective by Lemma 5.3,

$$\widehat{a}(\widehat{u}) = \widehat{a}(\widehat{v}) \iff e\big(q(\widehat{u})\big) = e\big(q(\widehat{v})\big) \iff q(\widehat{u}) = q(\widehat{v}),$$

which proves (4), and (5) is the same statement for kernel pairs.

The standard fiber classification for the map from an H_s-cover to a K_{ref}-cover identifies its based fiber with the right-coset set $H_s\backslash K_{\mathrm{ref}}$, and the inversion convention $H_s k \mapsto k^{-1}H_s$ identifies that pointed set with K_{ref}/H_s, proving (6). The map is one-sheeted exactly when the two subgroups agree, proving (7); and in finite index its sheet number is $[K_{\mathrm{ref}} : H_s]$, proving (8). □

5.3. **The defect at the canonical seed.**

Corollary 5.6 (Exact meaning of the forced collision)**.** *For an ordinary primitive canonical localization, Theorem 4.5, through display* (4.7) *of Theorem 4.4, gives*

$$H_s \subsetneq K_{\mathrm{ref}} = (i^p)_*^{-1}(H_t). \tag{5.16}$$

Consequently q is a nontrivial covering and e is an embedded resolved pullback component. The peripheral loop around the excess component gives a non-base point of the based fiber (5.14)*. All point collisions of the localized lift $\widehat{a}$ occur in fibers of this same q; none is left unexplained by the factorization.*

Proof. The proper inclusion (5.16) is (4.7), supplied at the ordinary primitive canonical seed by Theorem 4.4. By Theorem 5.5(7) the map q is injective if and only if $K_{\mathrm{ref}} = H_s$, so (5.16) makes q a noninjective covering, while Lemma 5.3 exhibits e as a topological embedding of the resolved pullback component. The peripheral class $[\gamma]$ supplied by Theorem 4.4 lies in $K_{\mathrm{ref}} \setminus H_s$, so its coset $[\gamma]H_s$ is a non-base point of (5.14). Finally Theorem 5.5(4)–(5) identify the kernel pair of $\widehat{a}$ with that of q, so every point collision of $\widehat{a}$ occurs inside a single q-fiber. □

Remark 5.7. Corollary 5.6 is unconditional at the canonical primitive seed: its hypotheses are supplied by Theorem 4.5 itself. The corresponding statement at a named Kahn–Lyubich decoration depth carries the additional stage-alignment hypothesis of the sequel.

Corollary 5.8 (The canonical sheet defect is infinite)**.** *At every ordinary primitive canonical seed,*

$$[K_{\mathrm{ref}} : H_s] = \infty. \tag{5.17}$$

Consequently q is an infinite-sheeted cover, and $\widehat{a}^{-1}\big(e(z)\big) = q^{-1}(z)$ is infinite for every $z \in P_{\mathrm{ref}}$. Equivalently: every non-singleton fiber of the completed localized inclusion — that is, every fiber over the embedded resolved complement component $e(P_{\mathrm{ref}})$ — is infinite, while fibers over the filled part are singletons and are not at issue. In particular no finite enumeration of the localization sheets can be complete.

Proof. Instantiate Definition 5.1 at the canonical seed, so that $X_s = \Omega_s$, $X_t = \Omega_t$, $a = i^p$ and $x_s = z_*$, and use the excess component X, the point $x \in X$ and the peripheral class $[\gamma]$ of Proposition 4.1 and Theorem 4.4. Winding about x gives a homomorphism $\mathrm{Ind}_x \colon \pi_1(\Omega_s, z_*) \to \mathbb{Z}$ with $\mathrm{Ind}_x(H_s) = \{0\}$ by (4.8), while (4.9) and (4.10) give $\mathrm{Ind}_x([\gamma]) = 1$ and $[\gamma] \in K_{\mathrm{ref}}$. Because K_{ref} is a subgroup, $[\gamma]^n \in K_{\mathrm{ref}}$ for every $n \in \mathbb{Z}$; and if $[\gamma]^m H_s = [\gamma]^n H_s$ then $[\gamma]^{m-n} \in H_s$, so applying Ind_x would give $m - n = 0$. The cosets $[\gamma]^n H_s$, $n \in \mathbb{Z}$, are therefore pairwise distinct, proving (5.17).

Theorem 5.5 identifies the fiber cardinality of q with this index and the $\widehat{a}$-fibers with the q-fibers, which gives the remaining assertions; the statement about the completed localized inclusion follows because, by Lemma 2.7, the filled completion identifies no two complement-cover points and adds only points of the marked continuum, whose fibers are singletons. □

Remark 5.9. Here $K_{\mathrm{ref}}/H_s = a_*^{-1}(H_t)/H_s$ is a pointed set of left cosets; it is not asserted to be a quotient group unless normality of H_s in K_{ref} is separately proved. Any quotient identifying these cosets requires an additional theorem showing that all later observables are constant on its fibers, and no statement below assumes such an identification.

5.4. **Uniqueness and transport.**

Theorem 5.10 (Unique embedded intermediate factor)**.** *Suppose $r_C \colon C \to X_s$ is a connected based intermediate cover and*

$$\widehat{X}_s \xrightarrow{q_C} C \xrightarrow{e_C} \widehat{X}_t \tag{5.18}$$

satisfies, with q_C and e_C based,

$$r_C q_C = p_s, \qquad p_t e_C = a\, r_C, \qquad \widehat{a} = e_C q_C, \tag{5.19}$$

and e_C is injective. Then the subgroup of $C \to X_s$ is exactly K_{ref}, and there is a unique based isomorphism of covers

$$C \cong P_{\mathrm{ref}} \tag{5.20}$$

identifying (5.18) *with the canonical factorization $\widehat{X}_s \xrightarrow{q} P_{\mathrm{ref}} \xrightarrow{e} \widehat{X}_t$.*

Proof. Let $L = (r_C)_* \pi_1(C) \le \pi_1(X_s)$. Existence of the based map q_C over X_s gives $H_s \le L$, and existence of e_C with $p_t e_C = a\, r_C$ gives $a_*(L) \le H_t$, so $L \le K_{\mathrm{ref}}$. Apply the subgroup-reflection criterion (Theorem 3.1, for the covers with subgroups L and H_t) to the lift $e_C \colon C \to \widehat{X}_t$ of the embedding a: since e_C is injective, that criterion gives $a_*^{-1}(H_t) = L$, whose left-hand side is K_{ref}; hence $L = K_{\mathrm{ref}}$. The classification of connected based covers

now gives the unique based isomorphism (5.20), and uniqueness of based lifts identifies both factor maps with q and e. □

Remark 5.11. Thus P_{ref} is the unique connected intermediate cover on which the remaining leg is embedded. The factor K_{ref}/H_s cannot be replaced by a finite label set without an additional theorem showing that all later observables factor through that quotient; and by Corollary 5.8 no finite label set is a bijective substitute at a canonical seed.

Proposition 5.12 (Exact sheet transport)**.** *For a path $\delta\colon [0,1] \to P_{\mathrm{ref}}$ and $\widehat{u} \in q^{-1}(\delta(0))$, let $\widetilde{\delta}_{\widehat{u}}$ be the unique q-lift of δ beginning at $\widehat{u}$ and set*

$$M_\delta(\widehat{u}) = \widetilde{\delta}_{\widehat{u}}(1) \in q^{-1}(\delta(1)). \tag{5.21}$$

Then M_δ is a bijection with inverse $M_{\delta^{-1}}$; it depends only on the homotopy class of δ relative to its endpoints; the constant path induces the identity; for composable paths

$$M_{\delta_1 * \delta_2} = M_{\delta_2} \circ M_{\delta_1}; \tag{5.22}$$

and at the based point these transports give the monodromy action on the pointed fiber K_{ref}/H_s. Hence

$$\Pi_1(P_{\mathrm{ref}}) \longrightarrow \mathbf{Set}_{\mathrm{bij}}, \qquad z \longmapsto q^{-1}(z), \quad [\delta] \longmapsto M_\delta \tag{5.23}$$

is a functor from the fundamental groupoid of P_{ref} to the category of sets and bijections. Its category of elements is the canonical sheet-transport groupoid $\mathcal{G}_q$, whose full object set $\{(z,\widehat{u}) : z \in P_{\mathrm{ref}},\ \widehat{u} \in q^{-1}(z)\}$ is canonically the point set of $\widehat{X}_s$; the pointed coset set K_{ref}/H_s is only its based fiber, equivalently the object set after restricting (5.23) *to the based action groupoid.*

Proof. Every assertion is forced by uniqueness and homotopy invariance of lifts for the covering q of Theorem 5.5(2). Lifting δ and then δ^{-1} returns to the initial point, which gives bijectivity. A homotopy relative to endpoints lifts once an initial sheet is fixed, so the terminal sheet is constant through the homotopy. Constant paths lift to constant paths. The lift of a concatenation is the concatenation of the two successive lifts; since $\delta_1 * \delta_2$ traverses δ_1 first, the first-traversed factor acts first and therefore appears innermost in the function composition, which is (5.22). Restricting (5.23) to the based fundamental group gives the usual monodromy action on the cover fiber, identified with K_{ref}/H_s by the inversion convention. □

Remark 5.13. No regularity of the cover is used. If H_s is not normal in K_{ref}, the based fiber K_{ref}/H_s is not a quotient group, but the groupoid (5.23) still gives complete path transport: under the inversion convention, continuation by a based loop g acts by the explicit formula (5.2), $k^{-1}H_s \mapsto g^{-1}k^{-1}H_s$, and not by a silently assumed quotient-group multiplication.

6. Kernel normal form and the winding quotient

Throughout this section we work at an ordinary primitive canonical seed with the shared local neighborhood and based representatives of Definition 2.6, and we write

$$a = i^p \colon \Omega_s \hookrightarrow \Omega_t, \qquad N_{\mathrm{fill}} := \ker\big[a_* \colon \pi_1(\Omega_s) \to \pi_1(\Omega_t)\big], \qquad K_{\mathrm{ref}} = a_*^{-1}(H_t).$$

We call a_* the *hole-filling homomorphism*: it is induced by the inclusion that fills back into the target every component of $F^{-1}B \setminus B$ removed from the source.

6.1. The sheet fiber is the hole-filling kernel.

Theorem 6.1 (Kernel normal form)**.** *At an ordinary primitive canonical seed, the restriction $a_*|_{H_s} \colon H_s \to H_t$ is an isomorphism. Consequently*

$$K_{\mathrm{ref}} = N_{\mathrm{fill}} \rtimes H_s,$$

and the map

$$N_{\mathrm{fill}} \longrightarrow K_{\mathrm{ref}}/H_s, \qquad n \longmapsto nH_s,$$

is a canonical pointed bijection.

Proof. By Lemma 2.5, $D_A \setminus A$ is an annulus; let c be its positively oriented core generator, so $\pi_1(D_A \setminus A) \cong \mathbb{Z}$ is generated by $[c]$. The image of $[c]$ in $\pi_1(\Omega_s)$, and likewise in $\pi_1(\Omega_t)$, has winding number one about every point of A; since the winding number about a fixed point of A is a homomorphism to $\mathbb{Z}$, the image of $[c]^n$ is nontrivial for every $n \neq 0$. Hence both induced maps

$$\pi_1(D_A \setminus A) \cong \mathbb{Z} \longrightarrow H_s, \qquad \pi_1(D_A \setminus A) \cong \mathbb{Z} \longrightarrow H_t$$

are isomorphisms onto their images. Because a restricts to the identity on the common physical neighborhood $D_A \setminus A$, the triangle formed by these two maps and $a_*|_{H_s}$ commutes, and $a_*|_{H_s} \colon H_s \to H_t$ is an isomorphism.

Let $k \in K_{\mathrm{ref}}$. Since $a_*(k) \in H_t$, there is a unique $h \in H_s$ with $a_*(h) = a_*(k)$; then $n := kh^{-1} \in \ker a_* = N_{\mathrm{fill}}$ and $k = nh$. Injectivity of $a_*|_{H_s}$ gives $N_{\mathrm{fill}} \cap H_s = \{1\}$, and N_{fill}, being the kernel of a_* on $\pi_1(\Omega_s)$, is normal in K_{ref}. Hence $K_{\mathrm{ref}} = N_{\mathrm{fill}} \rtimes H_s$. Finally, $k = nh$ gives $kH_s = nH_s$, and if $nH_s = n'H_s$ with $n, n' \in N_{\mathrm{fill}}$ then $n^{-1}n' \in N_{\mathrm{fill}} \cap H_s = \{1\}$; so $n \mapsto nH_s$ is a bijection onto K_{ref}/H_s, pointed because $1 \mapsto H_s$. □

Remark 6.2. The pointed coset set K_{ref}/H_s of Theorem 5.5 is not in general a group. Theorem 6.1 shows that at the ordinary canonical seed, with the shared based representatives of Definition 2.6, it nevertheless carries a canonical group model: the kernel of the hole-filling homomorphism. In one sentence, *canonical localization forgets exactly the kernel of the source-to-target hole-filling map.* The pointed-coset discipline of Section 5 remains the correct general-position language: the group model is a feature of the canonical seed.

6.2. **The exact excess count.**

Theorem 6.3 (Exact excess count)**.** *Let $f(z) = z^2 + c$ be primitively renormalizable of period $p > 1$ with ordinary return $F = f^p \colon U^p \to U^0$ and central continuum $A = K_p$. Then*

$$F^{-1}(A) = A \sqcup X_1 \sqcup \cdots \sqcup X_{2^p-2},$$

where each X_j is a full continuum disjoint from B, and $F|_{X_j} \colon X_j \to A$ is a homeomorphism.

Proof. Differentiating the composition,

$$F'(z) = \prod_{r=0}^{p-1} f'\big(f^r(z)\big) = 2^p \prod_{r=0}^{p-1} f^r(z),$$

so z is a critical point of F exactly when $f^r(z) = 0$ for some $0 \le r < p$, with critical value $F(z) = f^{p-r}\big(f^r(z)\big) = f^{p-r}(0)$. Since $0 \in A$ and $f^j(A) \subset K_j$, we get $f^{p-r}(0) \in K_{p-r}$. For $1 \le r < p$ the index $p - r$ lies in $\{1, \dots, p-1\}$, so by primitive disjointness the critical value lies in a member of $B \setminus A$ and in particular not in A. For $r = 0$ the critical point is $z = 0$ itself, with critical value $f^p(0) \in A$. Hence *the only critical point of F whose critical value lies in A is* 0.

Step 1: isolate A in the source. By Corollary 3.5 the preimage $F^{-1}(A)$ has finitely many components, each a full continuum. Choose a Jordan neighborhood $N_A \Subset U^p$ of A whose closure misses every component of $F^{-1}(A)$ other than A, with $\partial N_A \cap F^{-1}(A) = \varnothing$. Then $F(\partial N_A)$ is compact and disjoint from A.

Step 2: choose the target neighborhood. Choose a Jordan neighborhood $V \supset A$ with $\overline{V} \subset U^0$ whose *closure* avoids the three relevant compact sets:

$$\overline{V} \cap \Big((B \setminus A) \ \cup \ \big\{ f^{p-r}(0) : 1 \le r < p \big\} \ \cup \ F(\partial N_A) \Big) = \varnothing.$$

Shrinking V further if necessary, we may and do take $\overline{V}$ inside the target domain of the central degree-two polynomial-like restriction of F at A, so that the component V_0 of $F^{-1}(V)$ containing A maps properly onto V with degree exactly 2; this is the defining property of the primitive quadratic-like renormalization of period p and is what the degree budget below consumes.

Step 3: the central component. Let V_0 be the component of $F^{-1}(V)$ containing A (hence containing 0). Since V_0 is connected, meets N_A, and satisfies $F(V_0) \subset V$, while $\overline{V} \cap F(\partial N_A) = \varnothing$ by Step 2, the set V_0 cannot meet ∂N_A. By properness of F the closure $\overline{V_0}$ lies in $F^{-1}(\overline{V})$, which is likewise disjoint from ∂N_A; hence $V_0 \Subset N_A$. Every critical point of F lying in $F^{-1}(V)$ has critical value in V, hence equals 0; so V_0 is the unique component containing a critical point, and by the choice of V in Step 2 the restriction $F|_{V_0} \colon V_0 \to V$ is proper of degree exactly 2. Because N_A contains no component of $F^{-1}(A)$ other than A,

$$F^{-1}(A) \cap V_0 \subset F^{-1}(A) \cap N_A = A, \qquad \text{so} \qquad F^{-1}(A) \cap V_0 = A.$$

Step 4: the univalent components. Every other component V_j of $F^{-1}(V)$ contains no critical point, so it carries a proper unbranched covering of the simply connected domain V, hence a biholomorphism. The degrees over V sum to $\deg F = 2^p$, so there are exactly $2^p - 2$ biholomorphic components $V_1, \dots, V_{2^p-2}$.

Step 5: the excess continua. For $j \geq 1$ set $X_j = (F|_{V_j})^{-1}(A)$: a homeomorphic copy of A, compact, connected, and a component of $F^{-1}(A)$.

Step 6: fullness. Each X_j is a component of the p-fold pullback of the full continuum A, so it is full by Corollary 3.5.

Step 7: disjointness from B. This is the argument of Proposition 4.1: an intersection $X_j \cap K_l$ would map into $A \cap K_l = \varnothing$ for $l \neq 0$ by primitivity, while $X_j \cap A = \varnothing$ because distinct components of $F^{-1}(A)$ are disjoint. □

Corollary 6.4 (Exponential homological defect)**.** *By Corollary 3.5 the set $F^{-1}(B)$ has finitely many components, all full continua, among them A and the excess continua X_j. Apply Corollary 3.6 to this* entire *finite disjoint family and retain the disks belonging to the X_j: this yields pairwise disjoint isolating Jordan disks $D_j \supset X_j$ contained in U^p with*

$$D_j \cap \big(F^{-1}(B) \setminus X_j\big) = \varnothing,$$

so that $\partial D_j \subset \Omega_s$. Moreover $D_j \cap B = \varnothing$: a point of $D_j \cap B$ would lie in $B \cap U^p \subset F^{-1}(B)$ by Proposition 4.1(1), hence, as $X_j \cap B = \varnothing$, in $F^{-1}(B) \setminus X_j$, contradicting the displayed isolation. Choose based paths $\eta_j \subset \Omega_s$, peripheral loops $\gamma_j = \eta_j * \partial D_j * \eta_j^{-1}$, and points $x_j \in X_j$. Then each γ_j represents a class of $N_{\mathrm{fill}} = \ker a_*$, and the winding vector

$$W = \big(\mathrm{Ind}_{x_1}, \dots, \mathrm{Ind}_{x_{2^p-2}}\big) \colon N_{\mathrm{fill}} \longrightarrow \mathbb{Z}^{\,2^p-2}$$

is a surjective homomorphism with $W([\gamma_j]) = e_j$. Consequently

$$\operatorname{rank} H_1\big(N_{\mathrm{fill}}; \mathbb{Z}\big) \;\geq\; 2^p - 2,$$

and, via Theorem 6.1, the canonical sheet fiber K_{ref}/H_s carries at least $2^p - 2$ independent peripheral winding coordinates.

Proof. Each x_j lies in $F^{-1}B$, so $x_j \notin \Omega_s$ and Ind_{x_j} is a well-defined homomorphism $\pi_1(\Omega_s) \to \mathbb{Z}$. Each D_j is disjoint from B, hence contained in Ω_t, and γ_j contracts across D_j there, so $a_*[\gamma_j] = 1$ and $[\gamma_j] \in N_{\mathrm{fill}}$. Pairwise disjointness of the disks gives $\mathrm{Ind}_{x_i}([\gamma_j]) = \delta_{ij}$, i.e. $W([\gamma_j]) = e_j$; hence W is surjective. A surjection of N_{fill} onto the free abelian group of rank $2^p - 2$ factors through $H_1(N_{\mathrm{fill}}; \mathbb{Z})$, giving the rank bound. □

Remark 6.5. Corollary 6.4 sharpens Corollary 5.8 from "the sheet defect is infinite" to "the sheet defect has at least $2^p - 2$ independent winding coordinates." Any finite bookkeeping of the canonical localization must therefore lose not one but exponentially many independent peripheral directions, at every ordinary primitive seed.

7. Exact base change

The canonical resolution of Section 5 is produced at a single seed, while Kahn's tower is built from that seed by iterated exact equalizers. This section shows that the resolution survives that operation without loss: the covering leg, the embedded leg, the kernel pair, and the cardinality of every fiber are all preserved by base change. Once the exact equalizer tower of Proposition 2.3(vi) is fixed, the argument is topological: it uses only the inherited immersion, properness, and common marked-preimage properties of its stages, quoted where they are used.

7.1. **Equalizer and complementary-locus conventions.** All spaces are Hausdorff. For maps $i\colon X \to Z$ and $f\colon Y \to Z$ the fiber product is $X \times_Z Y = \{(x,y) : i(x) = f(y)\}$, the *kernel pair* of $g\colon X \to Y$ is $X \times_Y X$, and a commutative square is *cartesian* when it is a pullback square.

We write $\mathbf{U}^r$ for the r-th stage of the exact Kahn equalizer tower, $A^r \subset \mathbf{U}^r$ for its marked full continuum, and $i_r, f_r\colon \mathbf{U}^{r+1} \to \mathbf{U}^r$ for the two stage maps, the localized inclusion and the localized dynamics. The tower is the one over the *localized* seed: $(i_0, f_0) = (\widehat{\imath}, \widehat{F})$, so that $\mathbf{U}^0 = T_A$, $\mathbf{U}^1 = T'_A$, $A^0 = \widehat{A}$, and $d = 2$ at an ordinary primitive canonical seed (Proposition 2.3(iii)–(iv)); the continuum written A^r here is the set written K^r in Proposition 2.3(vi). In particular $\Omega^0 = \widehat{\Omega}_t$ and $\Omega^1 = \widehat{\Omega}_s$. With this indexing,

$$\mathbf{U}^{r+2} = \big\{(x,y) \in \mathbf{U}^{r+1} \times \mathbf{U}^{r+1} : i_r(x) = f_r(y)\big\}, \qquad i_{r+1}(x,y) = y, \quad f_{r+1}(x,y) = x. \tag{7.1}$$

By Definition 2.1, i_r is an immersion, f_r is a proper branched cover of degree d, and the legs have the common inverse

$$A^{r+1} = i_r^{-1}(A^r) = f_r^{-1}(A^r). \tag{7.2}$$

The *complementary locus* at stage r is $\Omega^r = \mathbf{U}^r \setminus A^r$; every statement of this section is made there.

Proposition 7.1 (Base-change propagation through exact equalizers)**.** *Let $X \xrightarrow{i} Z \xleftarrow{f} Y$ be continuous, and let $i = e \circ q$ with $q\colon X \to P$ a covering and $e\colon P \to Z$ an embedding. Put $E = X \times_Z Y$, $R = P \times_Z Y$, and*

$$q'(x,y) = (q(x), y), \qquad e'(p,y) = y, \qquad i'(x,y) = y, \tag{7.3}$$

so that $q'\colon E \to R$, $e'\colon R \to Y$, $i'\colon E \to Y$. Then:

(1) *q' is the pullback covering of q along the first projection $R \to P$; in particular the square*

$$\begin{array}{ccc} E & \xrightarrow{q'} & R \\ \downarrow & & \downarrow \\ X & \xrightarrow{q} & P \end{array}$$

with vertical maps the first-coordinate projections is cartesian;

(2) *e' is a topological embedding;*

(3) $i' = e' \circ q'$;
(4) *i' and q' have the same kernel pair;*
(5) $(q')^{-1}(p, y) = \{(x, y) : q(x) = p\} \cong q^{-1}(p)$ *for every* $(p, y) \in R$;
(6) *if moreover e is* open *— equivalently, if $i = e \circ q$ is a local homeomorphism, that is, an immersion in the holomorphic category — then e' is an open embedding and i' is again a local homeomorphism.*

Infinite fibers therefore stay infinite under base change.

Proof. (1) A point of $X \times_P R$ is a pair $(x, (p, y))$ with $q(x) = p$ and $e(p) = f(y)$, that is, a pair (x, y) with $i(x) = e(q(x)) = f(y)$; so $X \times_P R = X \times_Z Y = E$ with q' the second projection, and the square is cartesian. A pullback of a covering is a covering, the preimage of an evenly covered neighborhood being evenly covered. It is surjective: given $(p, y) \in R$ and $x \in q^{-1}(p)$, one has $i(x) = e(p) = f(y)$, so $(x, y) \in E$ and $q'(x, y) = (p, y)$.

(2) If $e'(p_1, y_1) = e'(p_2, y_2)$ then $y_1 = y_2 = y$, and membership in R gives $e(p_1) = f(y) = e(p_2)$, whence $p_1 = p_2$ by injectivity of e. On the image $\{y : f(y) \in e(P)\}$ the inverse of e' is $y \mapsto (e^{-1}(f(y)), y)$, continuous because e^{-1} is continuous on $e(P)$.

(3) By (7.3), $e'(q'(x, y)) = e'(q(x), y) = y = i'(x, y)$.

(4) As e' is injective, $i'(u) = i'(v) \iff e'(q'(u)) = e'(q'(v)) \iff q'(u) = q'(v)$.

(5) Immediate from the definition of q'.

(6) If e is open then $e(P)$ is open and $i = e \circ q$, a composite of a covering and an open embedding, is a local homeomorphism; conversely a local homeomorphism i forces $e = i \circ$ (local inverse of q) to be open. By (2), e' is injective with image $f^{-1}(e(P))$, open in Y by continuity of f, so e' is an open embedding; by (3), i' is then again a composite of a covering and an open embedding. □

Remark 7.2 (Clause (6) re-supplies the pullback hypothesis). Clause (6) is essential. The identification of the set-theoretic equalizer with the pullback in the category of complex 1-manifolds is available only when one of the two legs is an *immersion* [Kah23, Lemma 7.8, p. 40], and the source exhibits the failure explicitly on the next page: common ramification of order $m > 1$ turns the set-theoretic equalizer into m disks glued at a point, while the true pullback has m components [Kah23, p. 41]. Clauses (1)–(5) alone give only that e' is a *topological* embedding, which does not by itself make i' open; clause (6) supplies the missing openness, so that the hypothesis of that identification is regenerated at the next stage. Kahn obtains the same propagation independently [Kah23, Lemmas 7.7 and 7.9, p. 40]; clause (6) is its base-change form, recorded so that Corollary 7.4 does not borrow it silently.

7.2. **Propagation through the tower.**

Lemma 7.3 (The complementary locus is closed under the equalizer)**.** *For every $r \geq 0$,*

$$\Omega^{r+2} = \big\{(x,y) \in \Omega^{r+1} \times \Omega^{r+1} : i_r(x) = f_r(y)\big\} = \Omega^{r+1} \times_{\Omega^r} \Omega^{r+1}; \quad (7.4)$$

moreover f_r restricts to a surjection $\Omega^{r+1} \to \Omega^r$ onto the connected complementary locus, and hence the first projection $\Omega^{r+2} \to \Omega^{r+1}$ is surjective.

Proof. By (7.2) at stage $r+1$ and $i_{r+1}(x,y) = y$, the marked continuum at stage $r+2$ is $A^{r+2} = i_{r+1}^{-1}(A^{r+1}) = \{(x,y) : y \in A^{r+1}\}$; removing it from (7.1) leaves the pairs with $y \in \Omega^{r+1}$. For such a pair $f_r(y) \notin A^r$, so $i_r(x) = f_r(y) \in \Omega^r$ and $x \notin i_r^{-1}(A^r) = A^{r+1}$. This is (7.4).

For surjectivity: f_r is a proper holomorphic branched cover of degree d onto the disk $\mathbf{U}^r$ (Definition 2.1; [Kah23, Definition 7.1(3), p. 36]), a clause inherited at every stage, since the leg of the pullback opposite an immersion is again proper of degree d [Kah23, Lemma 7.9, p. 40] and the pullback is again a degree-d ψ-polynomial-like map [Kah23, Lemma 7.11, p. 42]. A proper nonconstant holomorphic map of positive degree onto the *connected* disk $\mathbf{U}^r$ is surjective, its image being open by openness of nonconstant holomorphic maps and closed by properness. Given $v \in \Omega^r$, choose y with $f_r(y) = v$; then $y \notin f_r^{-1}(A^r) = A^{r+1}$ by (7.2), so $y \in \Omega^{r+1}$ and $f_r(\Omega^{r+1}) = \Omega^r$. The locus Ω^r is connected, being the complement of a full continuum in a disk (Lemma 3.7 with $N = 1$), so (7.4) is formed over a connected base. Finally $i_r(x) \in \Omega^r$ for $x \in \Omega^{r+1}$, so some $y \in \Omega^{r+1}$ has $f_r(y) = i_r(x)$ and $(x,y) \in \Omega^{r+2}$. □

Corollary 7.4 (Propagation through the Kahn equalizer tower)**.** *Suppose the restricted inclusion at stage r factors as $i_r = e_r \circ q_r$, with $q_r \colon \Omega^{r+1} \to P^r$ a covering and $e_r \colon P^r \to \Omega^r$ an embedding, and put*

$$P^{r+1} = P^r \times_{\Omega^r} \Omega^{r+1}, \qquad q_{r+1}(x,y) = (q_r(x), y), \qquad e_{r+1}(p,y) = y. \quad (7.5)$$

Then $i_{r+1} = e_{r+1} \circ q_{r+1}$, where q_{r+1} is the pullback covering of q_r, e_{r+1} is an open embedding, i_{r+1} is again an immersion, the kernel pairs of i_{r+1} and q_{r+1} agree, the projection $P^{r+1} \to P^r$ is surjective, and $q_{r+1}^{-1}(p,y) \cong q_r^{-1}(p)$ for every $(p,y) \in P^{r+1}$. The hypothesis holds at $r = 0$. By induction on r, the resolved factorization of Theorem 5.5 therefore propagates, with the cardinality of every fiber, through every stage of the exact Kahn equalizer tower on the complementary locus; at an ordinary primitive canonical seed every non-singleton fiber at every stage is infinite.

Proof. By (7.4), Proposition 7.1 applies with $X = Y = \Omega^{r+1}$, $Z = \Omega^r$, $i = i_r$, $f = f_r$, $q = q_r$, $e = e_r$; then $E = \Omega^{r+2}$, $R = P^{r+1}$, and (7.5) is (7.3). Clauses (1)–(5) give the covering, the embedding, the factorization, the equality of kernel pairs and the fiber statement at stage $r+1$. Clause (6) applies as well, since i_r is an immersion at every stage [Kah23, Lemmas 7.7 and 7.9, p. 40]; it makes e_{r+1} an open embedding and i_{r+1} again an immersion, which is exactly the hypothesis under which the set-theoretic equalizer at stage $r+2$ is the pullback in the category of complex 1-manifolds [Kah23, Lemma 7.8,

p. 40], as in Remark 7.2. For surjectivity of $P^{r+1} \to P^r$, run the last step of Lemma 7.3 with $e_r(p) \in \Omega^r$ in place of $i_r(x)$: every $p \in P^r$ acquires a $y \in \Omega^{r+1}$ with $f_r(y) = e_r(p)$, hence a point $(p, y) \in P^{r+1}$, so the propagation is nonvacuous over every point of P^r.

Base case $r = 0$. On the complementary locus the stage-zero map is the lift $\widehat{a}$ of Section 4: here $\Omega^1 = \widehat{\Omega}_s$, $\Omega^0 = \widehat{\Omega}_t$ and $i_0 = \widehat{a}$, whose factorization is Theorem 5.5 instantiated at the canonical seed, namely $\widehat{a} = e \circ q$ with $q\colon \widehat{\Omega}_s \to P_{\mathrm{ref}}$ a surjective covering and $e\colon P_{\mathrm{ref}} \hookrightarrow \widehat{\Omega}_t$ a topological embedding. So the hypothesis holds with $P^0 = P_{\mathrm{ref}}$, $q_0 = q$, $e_0 = e$. The based fiber is infinite by Corollary 5.8, and q is a covering with connected base, so every fiber of q is infinite; iterating transports the factorization and, by clause (5), the cardinality of every fiber to all later stages. □

Remark 7.5 (Scope of the propagation). The propagation acts on the complementary locus only: it does not extend the factorization across a filled or prime-end completion. The restriction is forced rather than convenient, since [Kah23, §7.3.2, p. 42] records that the inclusion is a homeomorphism of the filled sets at the first stage and hence at every stage; each i_r is injective on A^{r+1}, and no nontrivial covering factor could exist over A^r. All of the content is annular, which is where Theorem 4.5 places the collision.

The Kahn–Lyubich decoration-stage applications of this propagation are developed in a sequel under an explicit stage-alignment hypothesis.

8. Consequences and outlook

Three operational consequences summarize what changes in arguments that use canonical localization.

First, embedding preservation may not be assumed: any argument treating the canonical localization of an ordinary primitive seed as embedding-preserving is invalid, not occasionally but always (Theorem 4.5), and the exact test is the reflection equality $(i^p)_*^{-1}(H_t) = H_s$ of Theorem 3.1, which fails canonically.

Second, the correct canonical object is the pair $(e(P_{\mathrm{ref}}), q)$: the embedded intermediate cover *together with* the covering whose based fiber K_{ref}/H_s is in canonical pointed bijection with N_{fill} (Theorem 6.1). Forgetting q forgets the kernel of the hole-filling homomorphism, hence at least $2^p - 2$ independent peripheral directions (Corollary 6.4).

Third, finite bookkeeping requires a theorem: since every non-singleton fiber is infinite, any finite quotient of the sheet data must be accompanied by a proof that the observables used later are constant on its fibers.

Two directions are left open. The Kahn–Lyubich decoration stages, tile lifts, and the whole-tile obstruction are developed in a sequel under an explicit stage-alignment hypothesis. And the kernel itself invites a sharper description: with r the number of source-only components of $F^{-1}(B) \setminus B$, a finite-connectivity argument with compatible peripheral bases should identify $(i^p)_*$ with a retraction of free groups $F_{p+r} \to F_p$ and N_{fill} with a free

group of countably infinite rank, sharpening the rank bound of Corollary 6.4 to equality with $\aleph_0$; the required lemma is not yet written, and we do not claim it.

9. Visual audit atlas

The following plates restate three exact parts of the argument in visual form. They are illustrations and audit aids only. The displayed theorems, equations, and exact ancillary certificates—not the rendered views—carry the proof.

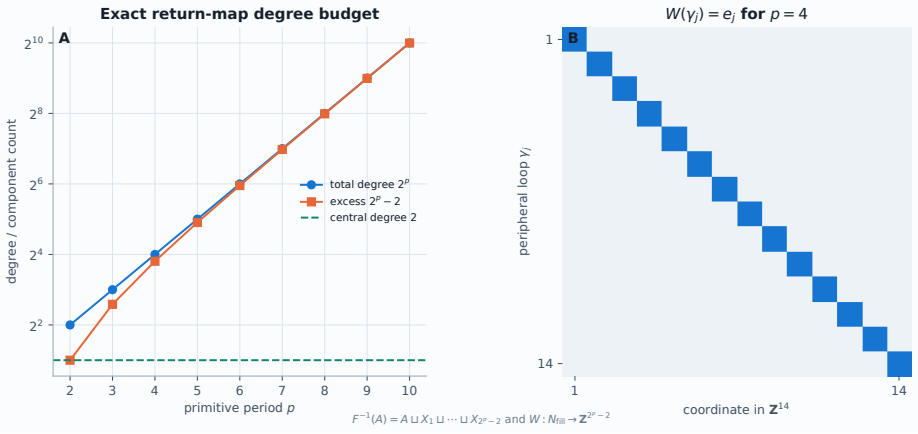


FIGURE 4. Visual audit of the degree budget in Theorem 6.3 and the winding quotient in Corollary 6.4. The exact identities are $\deg(F) = 2^p$, a central degree of 2, and therefore $2^p - 2$ excess components, together with the surjection $W\colon N_{\mathrm{fill}} \to \mathbb{Z}^{2^p-2}$ satisfying $W([\gamma_j]) = e_j$. The curves and marks illustrate these formulas; they are not numerical evidence and supply no additional proof step.

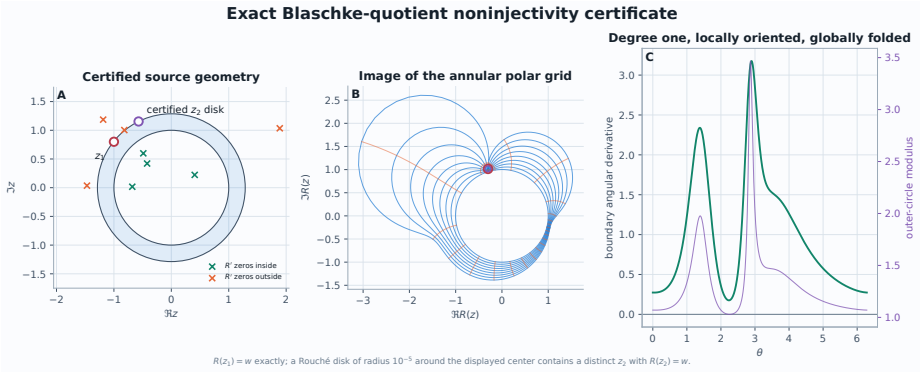


FIGURE 5. Visual audit of the exact Blaschke-quotient counterexample from Definition A.1, equation (A.1), and Theorem A.2. The source annulus has $r' = 1287/1000$, the certified point is $z_1 = -1 + \frac{4}{5}i$, and ancillary audit A7 in `supplement/certificates.json` gives an exact Rouché–Taylor disk containing a distinct z_2 with $R(z_2) = R(z_1)$. Decimal coordinates in the plate only render that exact certificate; the certificate and its validator, not the picture, establish the collision.

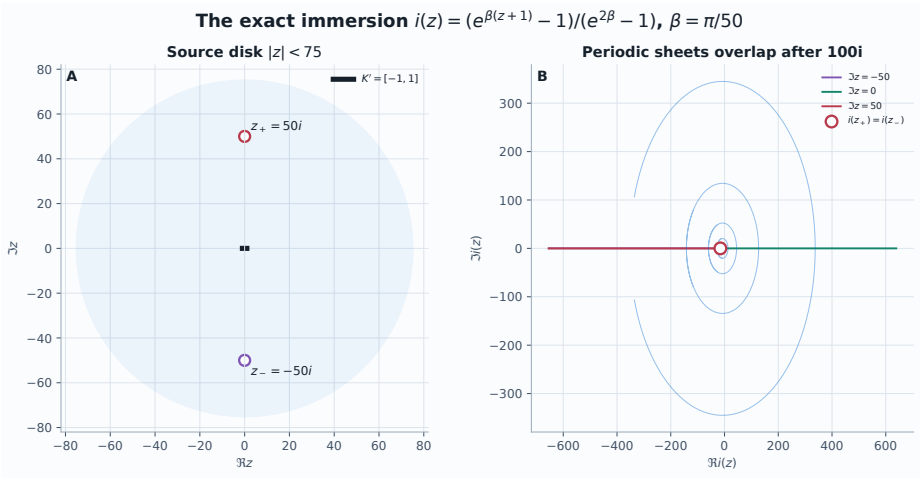


FIGURE 6. Visual audit of the abstract ψ-quadratic-like counterexample in Theorem A.3. Equation (A.2) sets $\beta = \pi/50$ and $i(z) = (e^{\beta(z+1)} - 1)/(e^{2\beta} - 1)$; the exact collision criterion (A.4) is $i(z) = i(w) \iff z - w = 100ik$. Thus $z_\pm = \pm 50i$ are distinct source points with $i(z_+) = i(z_-)$. The plotted sheets illustrate the periodic identity and are an audit aid, not a proof of it.

Appendix A. Two abstract counterexamples

The two examples of this appendix answer one question: can injectivity be recovered from the hypotheses the sources state alone, either in the annular form used for inner collars or in the full ψ-quadratic-like form of Definition 2.1? The answer is no in both settings. Both examples are abstract — neither is a dynamically generated Kahn or Kahn–Lyubich map — and both are compatible with the embedded inner collar of [Kah23, Theorem 7.3, p. 37] (cf. Remark 4.7).

Write $\mathbb{A}(1,\rho) = \{z : 1 < |z| < \rho\}$, $\mathbb{T} = \{|z| = 1\}$ and $\mathbb{D}_R = \{z : |z| < R\}$; for $|a| < 1$ the Blaschke factor is $B_a(z) = (z-a)/(1-\overline{a}z)$, and $\mathrm{wind}(\gamma, w)$ is the winding number of a loop γ about w. Two finite exact methods discharge every computational step below, and no floating-point arithmetic enters either. A *Sturm certificate* for $p \in \mathbb{Z}[t]$ is the exact Euclidean Sturm chain of (p, p') with its two leading-sign lists; Sturm's theorem returns the number of distinct real roots, and when that number is zero one rational evaluation fixes the sign of p on all of $\mathbb{R}$. A *Rouché certificate* for a polynomial P is a Gaussian-rational center c, a rational radius ρ, and the exact rational comparison on $|\zeta| = \rho$ of the linear Taylor term $P'(c)\zeta$ against $P(c)$ plus the higher Taylor tail, made through the modulus bounds $\max(|\Re w|, |\Im w|) \le |w| \le |\Re w| + |\Im w|$ for a Gaussian rational w; when it holds, Rouché's theorem places exactly one zero of P in $|z - c| < \rho$.

A.1. **A noninjective annulus immersion with degree-one analytic boundary.**

Definition A.1 (The Blaschke quotient R)**.** Let $r' = \frac{1287}{1000}$, $d = 16$, $r = (r')^{16}$, let

$$\mathcal{A} = \left\{ \tfrac{1+6i}{10},\ \tfrac{-3-i}{10},\ \tfrac{-7+2i}{10} \right\}, \qquad \mathcal{B} = \left\{ \tfrac{-6+2i}{10},\ \tfrac{4i}{10} \right\},$$

and put

$$R(z) = \frac{\prod_{a \in \mathcal{A}} B_a(z)}{\prod_{b \in \mathcal{B}} B_b(z)}. \tag{A.1}$$

Clearing denominators writes $R = N/D_0$ with N, D_0 of degree five and all coefficients Gaussian integers, so every sign, Sturm and Rouché computation below is a finite rational calculation.

Theorem A.2 (Annular no-go)**.** *With r', $d = 16$, $r = (r')^{16}$ and R as in Definition A.1, the map $R\colon \mathbb{A}(1, r') \to \mathbb{A}(1, r)$ is a holomorphic immersion, extends analytically across $\mathbb{T}$, and restricts to an orientation-preserving degree-one homeomorphism $\mathbb{T} \to \mathbb{T}$, with the exact modulus relation $r' = r^{1/d}$ built in; but R is not injective. Consequently the strengthening "a holomorphic immersion $I\colon \mathbb{A}(1, r^{1/d}) \to \mathbb{A}(1, r)$ that extends to the inner circle with degree one must be globally injective" is false, even when the inner-circle map is an orientation-preserving analytic homeomorphism.*

Proof. Five claims are verified; each is a finite exact computation, and the method that discharges it is named.

Claim 1 (zero and pole avoidance). R has neither zero nor pole on $1 \le |z| \le r'$, so $u = \log|R|$ is harmonic near that closed annulus. *Method:* the zeros and poles of (A.1) lie among $a, b, 1/\overline{a}, 1/\overline{b}$ with $a \in \mathcal{A}$, $b \in \mathcal{B}$; those points lie in $\mathbb{D}$, and exact squaring of $(r')^2 \max|a|^2$ and $(r')^2 \max|b|^2$ against 1 places every reciprocal point outside $|z| = r'$.

Claim 2 (the inner circle). $|R| \equiv 1$ on $\mathbb{T}$ and $\frac{d}{d\theta} \arg R(e^{i\theta}) > 0$ for every θ, so $R|_{\mathbb{T}}$ is an orientation-preserving degree-one homeomorphism. *Method:* each Blaschke factor has modulus one on $\mathbb{T}$, and the topological degree is the factor count $|\mathcal{A}| - |\mathcal{B}| = 3 - 2 = 1$. The Poisson-kernel identity $\frac{d}{d\theta} \arg B_a(e^{i\theta}) = (1 - |a|^2)/|e^{i\theta} - a|^2$ makes the angular derivative of R a difference of five positive terms; after the substitution $t = \tan(\theta/2)$ and clearing the positive Poisson denominators, its sign is that of an integer polynomial of degree ten, which an exact Sturm certificate shows to have no real root and whose value at $t = 0$ and leading coefficient are positive, the latter also covering $\theta = \pi$. A local orientation-preserving circle map of degree one is a homeomorphism.

Claim 3 (containment). $|R| > 1$ on $1 < |z| \le r'$ and $|R| < 48 < r$ there, so $R(\mathbb{A}(1, r')) \subset \mathbb{A}(1, r)$. *Method, lower bound:* parameterizing $|z| = r'$ rationally by $z(t) = r'(1 - t^2 + 2it)/(1 + t^2)$, exact substitution makes $|N|^2 - |D_0|^2$ a positive rational multiple of an integer polynomial of degree eight in t over a positive denominator; an exact Sturm certificate shows it has no real root, and its value at $t = 0$ and leading coefficient are positive. Since $u = \log|R|$ is harmonic near the closed annulus by Claim 1, vanishes on $\mathbb{T}$ by Claim 2 and is positive on $|z| = r'$, the minimum principle gives $u \ge 0$ and its strong form gives $|R| > 1$ on $1 < |z| < r'$. *Method, upper bound:* the exact maximum-modulus formula $\max_{|z|=\rho} |B_a(z)| = (\rho - s)/(1 - \rho s)$ for $s = |a| < 1$, $\rho s < 1$, monotone in ρ and in s, together with $|B_b| > 1$ outside $\mathbb{T}$, bounds $|R|$ by an explicit rational number, and two integer cross-multiplications give $|R| < 48 < (r')^{16} = r$. Finally $r = (r')^{16}$ by definition, so $r' = r^{1/d}$ identically.

Claim 4 (immersion). $R'(z) \neq 0$ for $1 < |z| < r'$. *Method:* up to a nonzero constant the numerator of R' is a degree-eight polynomial with Gaussian-integer coefficients. Eight exact Rouché certificates with one common rational radius isolate one zero apiece in eight disks about Gaussian-rational centers; exact squared-distance comparisons show the disks pairwise disjoint, and as the degree is eight they exhaust the zeros. Exact squaring classifies each disk as lying in $|z| < 1$ or in $|z| > r'$, four of each, so no zero lies in the source annulus; the denominator of R' is zero-free there by Claim 1.

Claim 5 (exact self-collision). For $z_1 = -1 + \frac{4}{5}i$ there is $z_2 \in \mathbb{A}(1, r')$, $z_2 \neq z_1$, with $R(z_2) = R(z_1)$. *Method:* $|z_1|^2 = \frac{41}{25}$ places z_1 in the annulus, and exact substitution gives $w = R(z_1)$ as a Gaussian rational. The quintic $N - wD_0$ has the exact factor $z - z_1$; one exact Rouché certificate locates a

zero z_2 of the remaining quartic in a small disk about a Gaussian-rational center, and exact squaring of that disk's rational radial endpoints places z_2 strictly inside the source annulus and away from z_1. As D_0 is zero-free there by Claim 1, z_2 is a genuine second solution of $R(z) = w$.

Claims 1–4 give the analytic extension, the boundary behavior, the containment with $r' = r^{1/d}$ and the immersion property; Claim 5 gives noninjectivity. □

The exact integer and rational witnesses for Claims 1–5 are supplied in the ancillary files `certificates.json` and `validate_certificates.py`, and can be rechecked by exact arithmetic alone: running

```
python validate_certificates.py
```

in the ancillary directory verifies all 144 exact checks of the seven audits and exits 0. The script depends only on `sympy`, uses integer, rational and Gaussian-rational arithmetic throughout, and refuses to run if any decimal literal is present in the witness file; no floating-point value enters any decision. It certifies arithmetic only: no topological or analytic step of this paper is machine-verified.

A.2. **An abstract ψ-quadratic-like pair with a remote collision.**

Theorem A.3 (ψ-quadratic-like no-go)**.** *Set $\beta = \pi/50$, $\Delta = e^{2\beta} - 1$, and*

$$\mathbf{U}' = \mathbb{D}_{75}, \quad \mathbf{U} = \mathbb{D}_{5625}, \quad f(z) = z^2, \quad K = [0,1], \quad K' = [-1,1], \quad i(z) = \frac{e^{\beta(z+1)} - 1}{\Delta}. \tag{A.2}$$

Then $(i, f)\colon (\mathbf{U}', K') \to (\mathbf{U}, K)$ is a degree-two ψ-polynomial-like map in the exact sense of Definition 2.1, and moreover

1. *$\mathbf{U}'$ and $\mathbf{U}$ are disks;*
2. *$K \Subset \mathbf{U}$ is a full continuum with more than one point;*
3. *$f\colon \mathbf{U}' \to \mathbf{U}$ is a proper holomorphic branched cover of degree two;*
4. *$i\colon \mathbf{U}' \to \mathbf{U}$ is a holomorphic immersion;*
5. *the common inverse $K' = i^{-1}(K) = f^{-1}(K)$ is connected;*
6. *$i|_{K'}\colon K' \to K$ is a homeomorphism;*
7. *the complementary-annulus map $i\colon \mathbf{U}' \setminus K' \to \mathbf{U} \setminus K$ induces $+1$ on fundamental groups; but*
8. *i is not injective, even on $\mathbf{U}' \setminus K'$.*

Consequently neither the proper degree-two companion map nor the common connected full-continuum pullback clause upgrades the immersion of an abstract ψ-quadratic-like pair to a global embedding.

Proof. The imaginary period of i is $2\pi i/\beta = 100i$: it exceeds the source radius 75, which excludes every nonzero logarithmic sheet from the preimage of K, but not the source diameter 150, which leaves room for a collision inside $\mathbf{U}'$. We take the clauses of Definition 2.1 in turn.

Disks and marked continuum. $\mathbb{D}_{75}$ and $\mathbb{D}_{5625}$ are simply connected plane domains; K and K' are compact, connected and nondegenerate, and a closed

segment does not separate the plane, so each is full, with $K \Subset \mathbf{U}$ and $K' \Subset \mathbf{U}'$. No local-connectivity assertion about a Julia set is used: these continua are explicit intervals.

Proper branched cover of degree two. $f(z) = z^2$ maps $\mathbb{D}_{75}$ onto $\mathbb{D}_{5625}$ with degree two counted with multiplicity, its only critical point being 0, of local degree two, with single branch value $0 \in K$. It is proper: for $C \Subset \mathbb{D}_{5625}$ pick $r_0 < 5625$ with $C \subset \overline{\mathbb{D}}_{r_0}$; then $f^{-1}(C) \subset \overline{\mathbb{D}}_{\sqrt{r_0}} \Subset \mathbb{D}_{75}$ is closed, hence compact.

Immersion into the target. Here $i'(z) = \beta e^{\beta(z+1)}/\Delta \neq 0$ on $\mathbf{U}'$, since $\beta, \Delta > 0$ and the exponential does not vanish. For containment without numerical approximation, $\Re(z+1) < 76$ gives $|i(z)| < (e^{76\beta}+1)/(e^{2\beta}-1)$, and the elementary bounds $3 < \pi < \frac{22}{7}$, $e < 3$, $e^x > 1 + x$ for $x > 0$ give $76\beta < 5$ and $e^{2\beta} - 1 > 2\beta > \frac{3}{25}$, whence $|i(z)| < \frac{6100}{3} < 5625$.

Common inverse. The condition $z^2 \geq 0$ forces z real and $z^2 \leq 1$ forces $|z| \leq 1$, so $f^{-1}(K) = K'$. For i, the equation $i(z) = t$ with $t \in [0,1]$ reads $e^{\beta(z+1)} = 1 + \Delta t$, whose right side is a positive real in $[1, e^{2\beta}]$; enumerating all complex logarithms gives

$$z = -1 + \frac{\log(1+\Delta t)}{\beta} + 100\, i\, k, \qquad k \in \mathbb{Z}. \tag{A.3}$$

As t runs over $[0,1]$ the real part in (A.3) runs over $[-1,1]$, while $k \neq 0$ forces $|\Im z| \geq 100 > 75$ and $z \notin \mathbf{U}'$. Hence $i^{-1}(K) = [-1,1] = K' = f^{-1}(K)$, connected. On K' the derivative i' is positive real with $i(-1) = 0$ and $i(1) = 1$, so $i|_{K'}$ is strictly increasing and a homeomorphism onto K, which is clause (6).

Collision criterion. From (A.2),

$$i(z) = i(w) \iff e^{\beta(z-w)} = 1 \iff z - w = 100\, i\, k \quad (k \in \mathbb{Z}). \tag{A.4}$$

Clause (7). Let $\gamma(\theta) = 2e^{i\theta}$, a positive generator of $\pi_1(\mathbf{U}' \setminus K')$; its image lies in $\mathbf{U} \setminus K$ because $i^{-1}(K) = K'$. The solutions of $i(z) = \frac{1}{2}$ are $z = x_0 + 100ik$ with $x_0 = -1 + \beta^{-1}\log(1 + \frac{\Delta}{2})$, and $1 < 1 + \frac{\Delta}{2} < e^{2\beta}$ gives $-1 < x_0 < 1$; so exactly one solution lies in $\mathbb{D}_2$, and it is simple. The argument principle gives $\operatorname{wind}(i \circ \gamma, \frac{1}{2}) = 1$, so the induced homomorphism of annular fundamental groups carries positive generator to positive generator. Degree $+1$ still does not imply injectivity.

Clause (8). Take $z_\pm = \pm 50i$: distinct, with $|z_\pm| = 50 < 75$, and outside K'. Since $\beta(z_+ - z_-) = 2\pi i$, criterion (A.4) gives $i(z_+) = i(z_-)$, and the common value $-(e^\beta + 1)/\Delta$ is negative, hence lies in $\mathbf{U} \setminus K$. So i has a genuine self-collision on the complementary annulus. □

Remark A.4 (Properness is assigned to the correct arrow). Definition 2.1 requires properness of f and not of i, and in this pair i is indeed not proper on the complementary annulus: reassigning that clause to i would conflate the two arrows.

Proposition A.5 (Compatibility with the local-embedding theorem)**.** *The pair (i, f) of Theorem A.3 contradicts nothing in* [Kah23, Theorem 7.3, p. 37]*: the map i is injective on every set of imaginary diameter less than* 100*, in particular on a sufficiently small neighborhood of $K' = [-1, 1]$. The local polynomial-like restriction and the embedded inner annular collar therefore remain valid for this pair; what fails is exactly the unsupported passage from the collar to the entire complementary annulus.*

Proof. By (A.4) every nontrivial collision has imaginary displacement at least 100, so i is injective on every set of imaginary diameter less than 100, including a small neighborhood of $[-1, 1]$. What [Kah23, Theorem 7.3, p. 37] supplies is a polynomial-like restriction: an embedding on a neighborhood selected near the filled set, from local injectivity there and truncation to an inner annular collar; [KL08, Lemma 3.1, p. 10] imports only that local embedding. The present pair meets both local conclusions: immersion yes, embedding near the filled set yes, global embedding no. □

Remark A.6 (Three verification points). Three points delimit what the examples establish. First, in Theorem A.2 the boundary-degree computation and the boundary-injectivity proof are independent: the degree is the factor count $|\mathcal{A}| - |\mathcal{B}|$, while injectivity on $\mathbb{T}$ comes from strict positivity of the angular derivative, a separate Sturm verification. Second, no floating-point root-finding is used anywhere: the existence of each located zero and the exhaustion of all zeros of a polynomial by disjoint disks both follow from exact rational Rouché inequalities together with a degree count. Third, the plane-case exception of the source — when the target is the whole plane, removability forces i to be an isomorphism [Kah23, Definition 7.1 and the following remark, p. 36] — is deliberately avoided, since Theorem A.3 uses the bounded hyperbolic disks $\mathbb{D}_{75}$ and $\mathbb{D}_{5625}$.

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